



Artificial Intelligence in *Materials by Design*: Critical Review And Perspectives on Materials Informatics to Generative and Agentic Intelligence

Kritesh Kumar Gupta¹ · Subrata Barman² · Anjana Sankar¹ · Sudip Dey² · Tanmoy Mukhopadhyay³

Received: 17 September 2025 / Revised: 11 November 2025 / Accepted: 14 December 2025 / Published online: 4 February 2026
© The Author(s) 2026

Abstract

For millennia, discovering new classes of materials has depended on intuition, experimentation, and often serendipity. However, recent technological advancements have established a more systematic, efficient and robust framework for materials design and discovery. Lately, there has been a surge in studies utilizing artificial intelligence (AI) to identify anomalies, predict stable compounds, material compositions and forecast effective properties. Such computational screening can help avoid unnecessary expenses, save time, and conserve resources during the initial stages of finding optimal solutions. With this motivation, the current review article consolidates research on the broad area of materials informatics, wherein the primary objective is to systematically highlight how the emerging AI frameworks are increasingly being exploited in materials design and discovery. It chronologically discusses online materials databases, data-driven design and optimization, machine learning (ML) and deep learning (DL) frameworks, computer vision-based approaches, natural language processing (NLP) in materials science, and transformer-based foundation models, along with generative and agentic intelligence. This review aims to present cutting-edge AI-driven frameworks in the context of materials science and engineering, providing the research community with critical insights into the progress, caveats and direction of research in this field.

1 Introduction

The introduction of new materials propels civilizations forward to the point where various periods of human history are frequently defined by the materials used in technology, such as the Stone, Bronze, and Iron Ages, and, more recently, the Silicon or Polymer Age. The design and discovery of novel classes of materials are crucial to technological growth because they make practical processes more efficient, dependable, and sustainable, allowing for breakthroughs that can handle difficult challenges across multiple industries [1]. For instance, the development of biocompatible

polymers and advanced ceramics has enabled advancements in targeted drug delivery systems and orthopaedic implants, resulting in substantial therapeutic breakthroughs [2–5].

Likewise, the discovery of lithium-based compounds [6] and perovskite materials [7] has facilitated the development of high-performance batteries and solar cells. Throughout history, military demands have frequently spurred material innovation, as seen in the development of advanced materials for weapons and protective equipment. The progress of composite materials and advancements in metallurgy can be linked to the need for enhanced performance in military applications [8, 9]. This highlights the multifaceted nature of material development, where the pursuit of innovations in health, sustainability, security, and efficiency fuels the search for new materials. The ongoing quest for superconductors (room-temperature) is a major focus of materials research, as such materials are promising in terms of preventing the energy losses associated with electrical resistance. The successful development of room-temperature superconductors could substantially help another ongoing research on materials for quantum computing by eliminating the requirement for costly and complex cooling systems,

✉ Tanmoy Mukhopadhyay
t.mukhopadhyay@soton.ac.uk

¹ Amrita School of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore, India

² Department of Mechanical Engineering, National Institute of Technology Silchar, Silchar, India

³ School of Engineering, University of Southampton, Southampton, UK

qubit stability, faster quantum operations, *etc.* [10]. With the recent technological advancements in computational capabilities, materials design and discovery have taken a route of artificial intelligence (AI) driven materials discovery. The rapid growth of machine learning (ML) and deep learning (DL) frameworks for materials characterization has transformed how materials are designed, discovered, and studied.

Trial-and-error experiments, which are time-consuming and resource-intensive, were a major part of materials discovery in the past. Furthermore, an Edisonian approach—where discoveries were made only after multiple failed attempts—was frequently the outcome of these methods. This is a result of the enormous number of possible combinations that arise from the intrinsic relationships between processing, structure, properties, and performance, as well as the complex nature of the materials design process [11–14]. This created a compelling case for using the computationally effective frameworks under the materials genome initiative's (MGI) jurisdiction to develop a strong mechanism for materials design [15]. The schematic of the materials design pipeline for the traditional approach versus materials-informatics is illustrated in Fig. 1(A). The effective use of materials-informatics frameworks has shown remarkable promise in exploring uncharted territory and creating new opportunities for the rapid and accurate discovery of materials. For example, a study by Raccuglia et al. [16] showed that the conditions for the best product formation were predicted by applying cheminformatics techniques to the unsuccessful hygrothermal syntheses.

With the motivation of exploring the current state-of-the-art in artificial intelligence-assisted materials design, the current review article puts together the critical aspects of materials informatics and its paradigm-shifting influence on the future course of materials design and discovery. The focus of this article is primarily on the recent efforts of the research community in the deployment of cutting-edge AI techniques in designing different classes of materials. In addition to that, the article captures the journey of materials-informatics concerning the exploitation of conventional machine learning models to transition into the potential success of utilising the capabilities of generative artificial intelligence (GenAI) and Agentic AI for optimal materials discovery. A general overview of machine learning is presented in Fig. 1(B), highlighting its tools, sub-fields, complementary disciplines, and problem categories. The article discusses different classes of materials, where the successful use of AI has made a significant contribution to the expedited design by revealing intricate insights about the material's performance. Throughout reviewing the exploitable nexus of AI and materials, we would specifically focus on the recently reported AI-driven design and discovery of three different classes of representative materials:

two-dimensional (2D) materials, composite materials, and high entropy alloys (HEA), to capture a wide spectrum of the trends in material innovation. In the following sections, we systematically discuss the evolution and adoption of AI in materials design and discovery, along with resources and tools.

2 Data Mining for Introducing AI in the Materials Domain

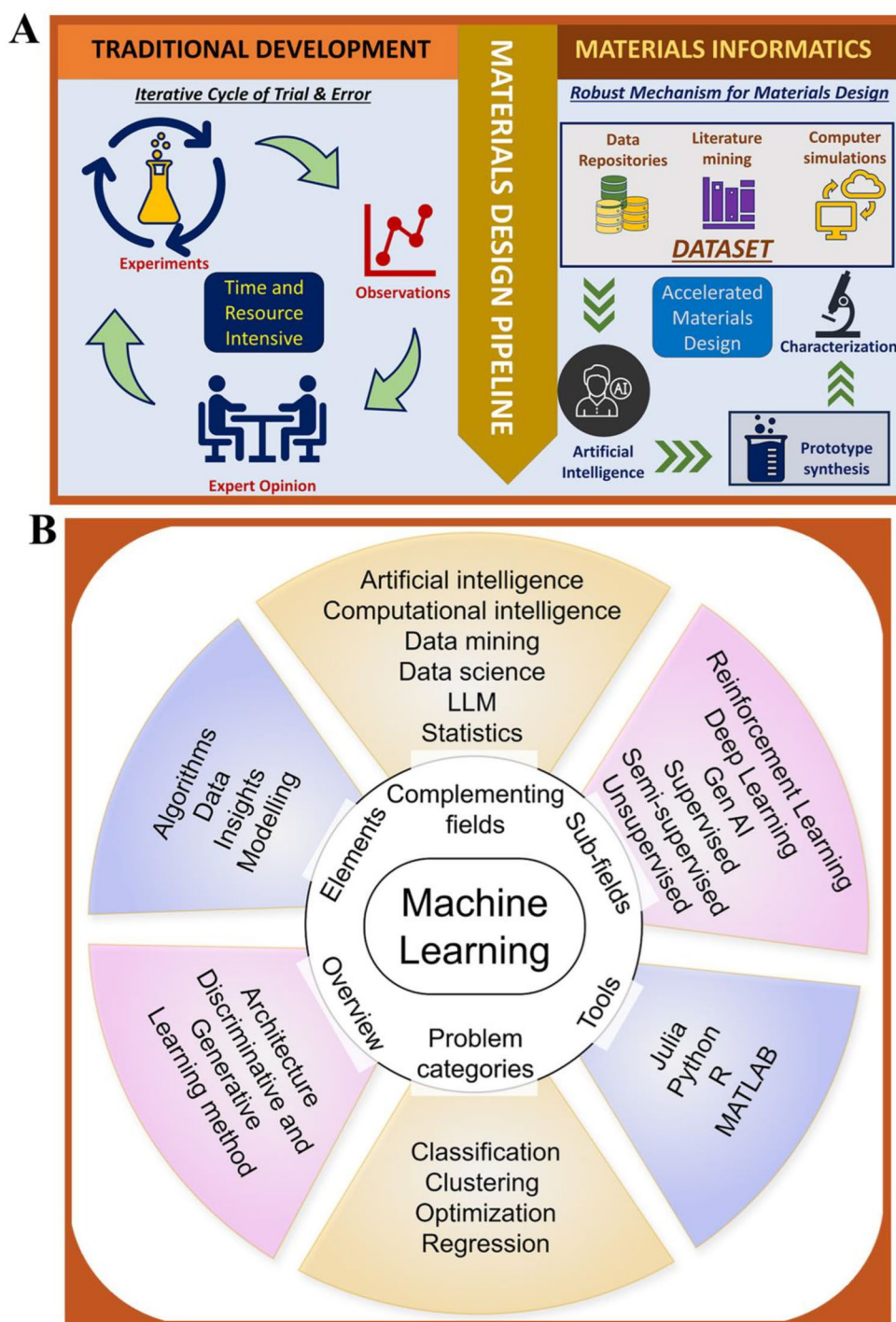
This section discusses various open-source data repositories that support the development of AI and ML models capable of handling multimodal information, including tabulated, CIF, JSON, and text-based datasets. It also highlights the importance of data standardization and integration strategies aimed at addressing the pervasive challenge of materials data heterogeneity, thereby enabling seamless interoperability and more reliable model development within the materials informatics framework.

2.1 Open-Source Data Repositories

The first step to advance in any informatics domain requires structured/unstructured information to start the investigation. In general, the materials research community prefers to generate the data by performing large-scale experiments/simulations due to obvious reasons (accuracy and reliability). However, most often, the constraint in terms of the lack of resources presents a major bottleneck in applying the AI-driven techniques. The curse of data unavailability (relevant) is a great challenge to start with, though the recent efforts of the materials research community are providing great help in mitigating this challenge. For instance, the open-source initiative by different publishing groups has provided a platform for researchers to publish their research data, separately. Journals such as *Data in Brief* (Elsevier), *Open Data* (Springer), *Data* (MDPI), *etc.* provide a wide range of domain-specific datasets useful for materials-informatics investigation. However, despite the abundance of datasets in these journals, a significant barrier persists in finding high-throughput datasets for specific material systems. The challenge stems from the resource-intensive nature of generating large-scale datasets. In the context of such challenges, the following paragraphs highlight some of the available open-source repositories which offer solutions for a wide range of materials systems.

The most popular comprehensive dataset is The Materials Project [17], which offers compositional, structural, and functional insights about different classes of materials systems such as catalysts, perovskites, battery materials, metal-organic frameworks, *etc.* The Materials Project (MP)

Fig. 1 Embedding materials domain knowledge into machine learning Algorithms. (A) A comparison between the AI/ML-driven design pipeline (right) and the conventional trial-and-error method for materials design (left). The conventional cycle entails conducting a large number of experiments that are verified by subject-matter specialists. In contrast, the materials-informatics pipeline leverages the predictive power of ml models combined with exploration strategies, such as genetic algorithms, Bayesian optimization, and knowledge graphs, to efficiently identify optimal solutions. In the latter, the experiments are conducted primarily during prototype development, based on AI/ML-suggested candidates, significantly reducing experimental costs and harnessing computational intelligence to emulate expert guidance. (B) Machine learning model development, learning overview, tools and application landscape



application programming interface (API) in the Python environment helps users to query the materials data easily, which is mostly used by materials researchers for developing several derivative frameworks. For instance, Ong et al. [18] developed the REpresentational State Transfer (REST) API to communicate with the Materials Project webpage for extracting the information from the materials database. Dung et al. [19] utilized the MP-API to develop

a computational framework for analysing the X-ray diffraction. Liu et al. [20] developed a machine-learning model by using the perovskite dataset available in MP-API to screen the stable ABO_3 perovskites. Haastrup et al. [21] performed large-scale density function theory (DFT) calculations for ≈ 2000 variations of two-dimensional (2D) materials to record their structural, electronic, and magnetic properties. The dataset is available openly as the Computational

2D Materials Database (C2DB). The crystallography open database (COD) by Gražulis et al. [22] contains the crystal structure information of more than 0.38 million organic, metal-organic materials. Similarly, the Open Quantum Materials Database (OQMD) by Saal et al. [23] consists of the crystal structure information of $\approx 200,000$ compounds produced by high-throughput density functional theory. The crystallographic information available in these open-source repositories has been successfully used by the research groups to develop machine learning (ML)/deep learning (DL) based predictive frameworks [24–26].

Apart from these open-source repositories, third-party open-source repositories such as Materials Project (MP), Citrine, and Materials Data Facility (MDF) are frequently queried for materials data using the Python-based open-source library MatMiner [27]. The use of MatMiner for materials data mining and subsequent critical material property prediction has garnered significant interest from the materials research community [28–31]. Another popular open-source library for materials data mining is Pymatgen [32], which is based on Python and allows users to create the input files needed to perform DFT calculations and query the Materials Project (MP) database for materials data. Large-scale AI-driven materials property screening and prediction were also made possible by the effective use of the Pymatgen library [33–35]. The MatMiner and Pymatgen library websites provide comprehensive documentation, including installation instructions and guidance on how to use them for materials mining, along with several example use cases. Data-driven materials design is now supported by a large

number of open repositories that consist of a variety of computational and experimental datasets. The frequently used materials databases, the kinds of content they offer, and the AI-driven models they can support are compiled in Table 1.

2.2 Integration and Standardization of Heterogeneous Materials Data

The preceding subsection lists the available open-source data repositories readily deployable for materials-informatics applications. While numerous databases are available, the extraction of relevant data mostly requires navigating the various sources, including the published literature. The effectiveness of the developed AI frameworks greatly relies on the consistent, continuous and high-quality dataset. Querying the heterogeneous dataset available at numerous sources presents extreme challenges in the development of a unified AI system due to the lack of integration across the inherent material features, such as composition, crystal structure, etc. Such challenges call for utilization of strategies for data and representation level integration.

Data-level integration: The data-level integration requires the use of standard data formats, such as CIF for structure data, JSON for materials metadata, SMILE/SELFIES for organic compounds, etc. The use of materials-specific application programming interface (APIs) such as Materials Project [17], PyMatgen [32], MatMiner [27] also facilitates the proper integration of materials datasets on the specific data-modality level. For instance, these APIs facilitate the querying of materials data on the basis of

Table 1 AI and machine learning applications created for predictive modeling, feature extraction, and knowledge discovery in materials science, along with a summary of the most popular repositories and databases in the materials informatics ecosystem

Repository/Database	Primary Content/Data Type	Typical Data Format	AI/ML Models in Materials-Informatics regime
Materials Project [17]	structure-property correlation data, composition-property correlation, electronic, mechanical, and thermodynamic properties	Tabular, JSON/CSV	AI/ML models for Tabulated dataset [36], developing materials ontology by using text data [37].
OQMD [23]	Database of DFT calculated thermodynamic and structural properties	SQL/CSV	Property prediction through AI/ML frameworks [38]
AFLOW [39]	Structural and crystallographic data, Thermodynamic data, electronic, mechanical, magnetic & thermal properties	JSON, Text, CIF	Interatomic force field developed [40]
NOMAD Repository [41]	Raw and processed DFT data from multiple codes (VASP, QE, etc.)	HDF5/JSON	High-throughput Calculation for solar energy conversion of perovskites [42]
Citrine/Citrine Platform [43]	Experimental and computational datasets for industrial materials	CSV	Machine Learning model for predictive framework [44]
JARVIS (NIST) [45]	DFT, ML, and experimental data; includes 2D materials and polymer datasets	JSON/CIF/Text	Graph neural networks for predicting electron density of states [46]
MatMiner [27]	Benchmark datasets for ML validation (regression & classification tasks)	CSV	Elemental descriptors generated for Lithium-ion battery [47]
MatScholar [48]	Text-mined materials science literature	Text (XML, JSON)	Named entity recognition for materials science [49]
C2DB (Computational 2D Materials Database) [21]	2D material structures and electronic properties	JSON/CIF	Predictive framework by using the machine learning models [50]

composition, structural configurations, and stability of the materials. The unified system for materials querying offers a proper integration strategy for a wide variety of materials systems.

Representation-level integration: Another strategy for data integration and standardization is facilitated by the representation of materials systems, for instance, the use of materials embeddings [51] for representing the compositional or structural formula helps in the development of a generalized AI framework capable of making a reasonable prediction for any new materials system. The use of materials ontology for developing a materials knowledge graph also provides a scalable framework for knowledge interpretation.

Despite the data and representation level integration, a challenge remains in processing the information obtained from the different modalities of the dataset, including textual excerpts of the published literature, images (microstructures, simulation frames, etc.), structural configurations (CIF files), etc. The development of AI frameworks to address these challenges represents a current and rapidly evolving direction of research. To address the integration of heterogeneous datasets, the recent cutting-edge models, such as LLM agents, offer a unified AI framework capable of processing the multi-modal information [52]. The detailed discussion on the Agentic AI approach is presented in Sect. 6.

3 Materials-By-Design: Materials Informatics

Materials-by-design is a new-generation approach which aims at accelerating the design and discovery of function-specific materials by algorithmically exploring a large sample space during computational screening. This approach leverages machine learning methods and theoretical calculations (Numerical simulations, Molecular Dynamics simulations, Density Functional Theory, etc.) to reduce the timeline for materials discovery from decades to months. The complex relationship between a material's processing, structure, and properties (PSP) drives its functional performance. Strategically modifying a material's attributes (such as structure and composition) at the nano- and microscale can enhance its performance [53–58]. Exploring materials with desired properties through manual experimentation is hindered by the infeasibility of large-scale experimentation (both physical experiments and physics-based simulations) needed to understand complex process-structure-property relationships. *Materials-by-Design* offers the solution in terms of implementing AI-integrated computational simulations for exploring and controlling the relationship between structure, processing, composition and properties.

Figure 2(A) illustrates a prospective end-to-end pipeline of the Materials-by-Design concept, wherein the *Materials Discovery* stage involves exploring an existing materials database by using the concepts of materials-informatics. The successful implementation of these concepts has shown promise in revealing the deep insights of PSP linkages, for instance, Liu et al. [59] deployed machine learning for establishing the relationship between processing, structure and mechanical performance of additively manufactured metal bars. Thoppil et al. [60] deployed a Bayesian framework to establish PSP linkages of complex concentrated alloys by exploring the thermo-mechanical performance dataset. Tran et al. [61] developed an ML-driven predictive framework for rapid characterization of polymers in terms of thermal, mechanical and electrical performance. The integration of AI/ML in materials research has shown great promise in terms of accelerated search for materials with desired performance. In the following sections, the generalized application of the concepts of materials informatics is demonstrated by reviewing the recently published literature covering a broad range of materials.

4 AI-Driven Materials Design and Discovery

This section presents an extensive review of the application of different AI algorithms for designing, discovering, and characterizing different classes of material systems. A general overview of the integration of AI concepts into various aspects of materials design is illustrated in Fig. 2(B). The studies discussed in this section, covering different classes of materials, highlight that with the successful implementation of AI-driven frameworks, a robust mechanism for materials design and discovery can be developed. The challenges of traditional materials discovery, such as the time and cost involved in screening for optimal solutions, can be effectively addressed by using an AI-driven framework.

4.1 Two-Dimensional (2D) Materials

The design of 2-dimensional materials (2D) requires nano-engineering of the atomic arrangement of the material system for programming their electronic, optical, mechanical and magnetic properties. For scalable synthesis of such materials, it is essential that mass-characterization is performed to find the desired solution. In this regard, Frey et al. [63] employed a deep learning framework for predicting the electronic properties of defective 2D materials. With this framework, the correlation between structural defects and the electronic properties of different 2D materials is developed, which is capable of predicting the bandgap and Fermi energy of any 2D material subjected to point defects.

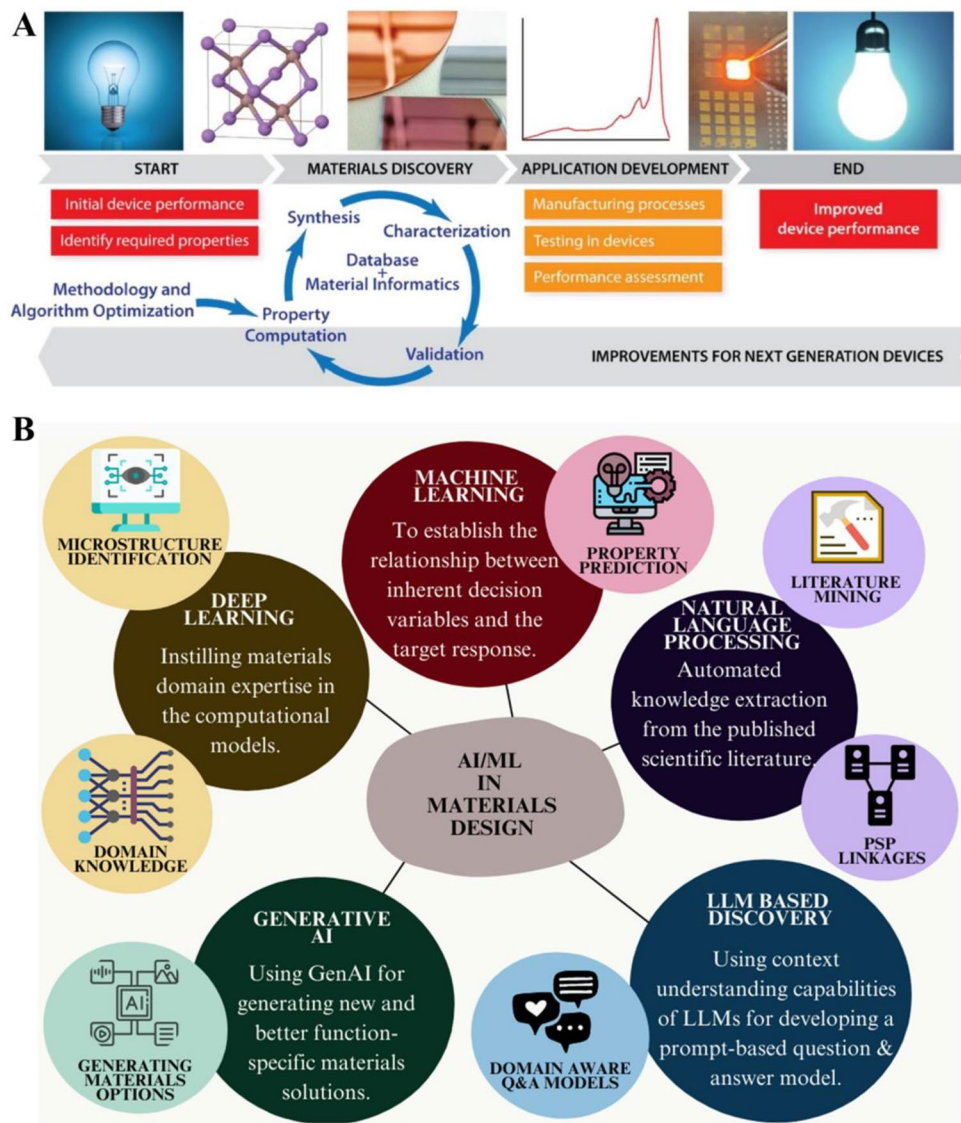


Fig. 2 Harnessing materials informatics for integrated design and fabrication of enhanced-performance Devices. **(A)** To assess important target properties, the process starts with initial device testing, which creates crucial baseline data for further analysis. In order to optimize processing conditions and internal material structures and achieve desired performance criteria, this data is combined with active learning techniques and embedded in machine learning-driven predictive models. The knowledge gathered from this data-driven investigation guides the creation of prototypes and the manufacturing of finished goods after the required validation procedures. The time and cost typically involved with trial-and-error screening techniques are greatly reduced by this simplified method (Reproduced from Alberi

et al. [62]). **(B)** Schematic showing how different aspects of artificial intelligence (AI) can be integrated into the materials design process. Conventional machine learning (ml) methods are used to identify relationships between key variables and material properties, helping predict unknown responses. Deep learning (dl) models can handle multi-modal data such as tables, images, and text to extract materials-specific knowledge. Generative AI supports inverse design by suggesting material systems that meet desired performance criteria. Large language models (LLMs) contribute by capturing domain knowledge and enabling intelligent Q&A systems, while natural language processing (NLP) tools help mine literature to uncover links among processing, structure, and property information

In another study by Frey et al. [64], an unsupervised ML approach is utilized to predict the likelihood of the synthesizability of a wide range of 2D MXenes based on the elements involved and first-principle calculations. Zhang et al. [50] used the open-source C2DB database for 2D materials to develop a machine learning-based framework to predict the bandgap of unknown 2D materials. Despite property

prediction, unsupervised ML frameworks are also used to reveal the intricate patterns in the epitaxial growth of 2D thin films. For instance, Kim et al. [65] utilized the sequential video from reflective high-energy electron diffraction data to unravel the growth modes of ReSe₂, wherein they utilized principal component analysis and K-means clustering to visualize the trends in the patterns of the epitaxial

growth of the 2D materials. From the perspective of ascertaining the stability of the synthesized 2D materials, Manti et al. [66] employed a classification model with the C2DB database to report the categories of 2D materials which possess zero-temperature dynamic stability. Similarly, Choudhary et al. [67] used classification ML models to report the bilayer heterostructures of the 2D materials having band alignments and lattice mismatch.

The promising potential of 2D materials in enhancing the operational performance of wearable devices, structural materials and a range of sensors makes it essential to keep track of their mechanical performance as well [68, 69]. The applied deformation should be within the permissible limits so that the operational performance of such devices is not affected. In this regard, to have comprehensive insight into the deformation capabilities of defective and doped graphene layers, Gupta et al. [70] reported an ML-driven predictive framework to ascertain the deployability of defective (point and Stone-Wales defects) and doped (Si-doped) graphene layers based on the fracture and elastic properties. They also considered temperature and strain rate as important features while developing the ML model (refer to Fig. 3). Other derivatives of graphene monolayers, in terms of twisted bilayer graphene, are found to be explored by integrating ML-driven frameworks with the molecular dynamics (MD) simulation to comprehensively characterize mechanical, vibration and impact capabilities offered by these 2D configurations of graphene [71–76]. To better

understand the atomic-scale stress and strain distribution in graphene monolayers, images from MD simulations (incorporating point defects and polycrystallinity) were used to train a deep learning framework based on convolutional neural networks. This framework can generate stress and strain fields based on the topological information regarding the presence of point defects in the graphene monolayer [77, 78]. Dewapriya et al. [79] developed a shallow and deep neural network to comprehensively assess the fracture response of monolayer graphene by considering temperature, strain rate, vacancy concentration, and edge configuration. The brief review presented in this subsection highlights some of the recent attempts to utilize the concepts of materials-informatics for optimal design and synthesis of 2D materials with the aim of enhancing operational performances.

4.2 Composite Materials

The quest for obtaining a high strength-to-weight ratio in structural materials piqued the interest of the materials community in harnessing the advantages offered by composite materials. The functional performance of composite materials depends on several factors such as the type of matrix (metal, ceramic or polymer), the type of reinforcement material (carbon fibre, glass fibre, nanomaterials like graphene, carbon nano-tubes, SiC particles), the processing method of the composite, and several other factors like operational conditions (temperature, humidity, corrosive

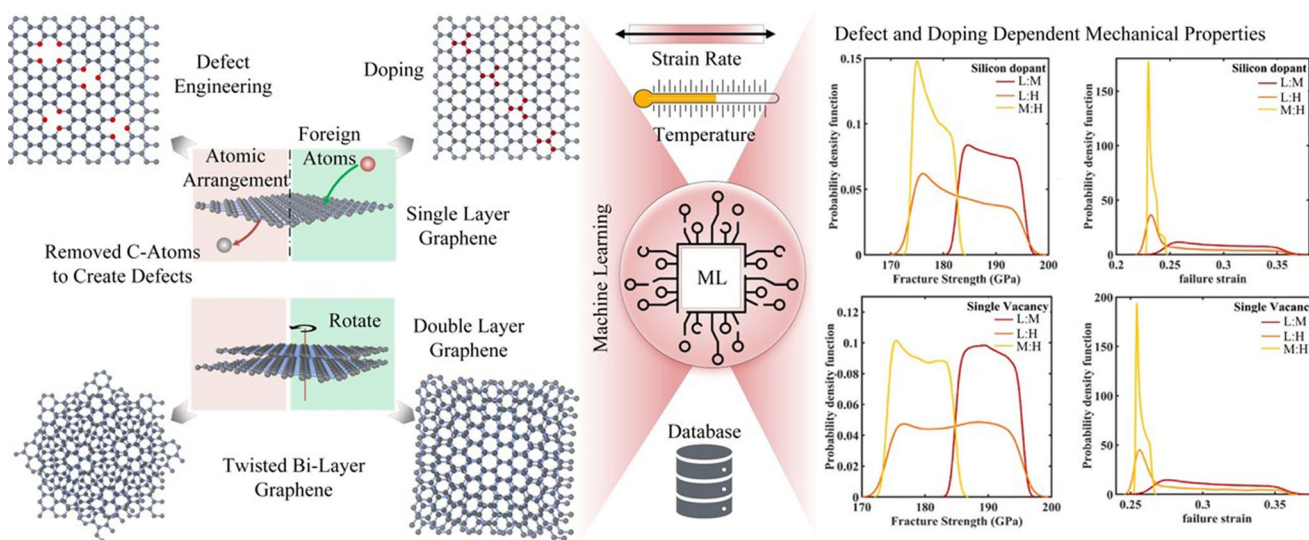


Fig. 3 Leveraging machine learning frameworks for atomistic characterisation of strategically doped and defective graphene monolayers. Atomic-scale engineering of two-dimensional materials like graphene enables exceptional functional performance but often comes at the cost of reduced mechanical strength. For a thorough understanding, it is essential to correlate nanoengineering features, such as the concentration and distribution of dopants and defects, with the resulting mechanical behaviour of graphene derivatives. To achieve this,

atomistic simulations using molecular dynamics are integrated with machine learning frameworks. The resulting predictive models capture how fracture properties vary in response to different configurations of vacancy defects and silicon dopants, under a range of temperatures and strain rates [70–76]. Note that ml can be exploited in a similar framework to quantify the effect of defect and doping for other mechanical and multi-physical properties of different 2D materials

environment, *etc.*) [80]. To design composites for function-specific criteria, it is essential to carefully select the combination of these factors. The traditional trial-and-error approach to finding the optimal solution for designing such composites is a time and resource-intensive endeavour. This highlights the need for AI/ML-driven strategies, which can accelerate the design process by simultaneously exploring a wide range of parametric combinations (refer to Fig. 4); something that would be difficult to achieve through traditional physics-based computational simulations or physical experiments [81].

To this end, a vast number of studies reported the development of ML-driven predictive frameworks. For instance, Raturi et al. [82] compared the predictive performance of different ML models for predicting the natural frequencies of functionally graded material (FGM) as a function of stochastic variation in the porosity. Similarly, Vaishali et al. [83] developed a support vector machine (SVM) based ML model to predict the vibration and impact response of an FGM plate subjected to stochastically varied material properties. In another work, Vaishali et al. [84] attempted to capture the influence of uncertainty in the materials and structural configurations of FGM shells by incorporating the SVM-based predictions of dynamic responses. Divakaraju et al. [85] used the computational capability of Gaussian process regression (GPR) to predict the mechanical properties of unidirectional (UD) carbon-epoxy composite. Yossef et al. [86] created the interpretable extreme gradient boosting (XGBoost) machine learning model, which is used in the design of fiber-reinforced plastics (FRP), to get around

the deployment problems that come with black-box ML models. The ML model's explainability made it easier to determine how each input parameter contributed to the prediction mechanism.

Insightful studies in this domain also include physics-informed neural networks (PINNs), which are highly effective at capturing the physics underlying the physical phenomenon, in addition to machine learning (ML)-based predictive frameworks. For example, Yao et al. [87] developed PINN to capture the growth of the fatigue crack in composite laminates. They utilized the experimental load history till Mode I failure to predict the crack growth possibilities. Similarly, Bharadwaja et al. [88] utilized the PINN framework for predicting the elastic deformation pattern of heterogeneous materials. They also captured the uncertainty in prediction by performing a comparison of the predicted pattern with the finite-elements (FE) based simulations. Zhang et al. [89] utilized the PINN framework for predicting the creep rupture life of ceramic matrix composites. Moving ahead in the complexities associated with the composite materials design, deep learning models have offered solutions to some of the complex challenges. For instance, Khan et al. [90] integrated the structural vibrations with the convolution neural network (CNN) for detecting the delamination in composite laminates - such a framework helps in the early detection of delamination. Yan et al. [91] reported a CNN-based framework applied to predict the crack morphology of UD composites. They utilized the FE simulation-based images for developing the CNN model. Zhou et al. [92] also utilized the CNN framework for capturing

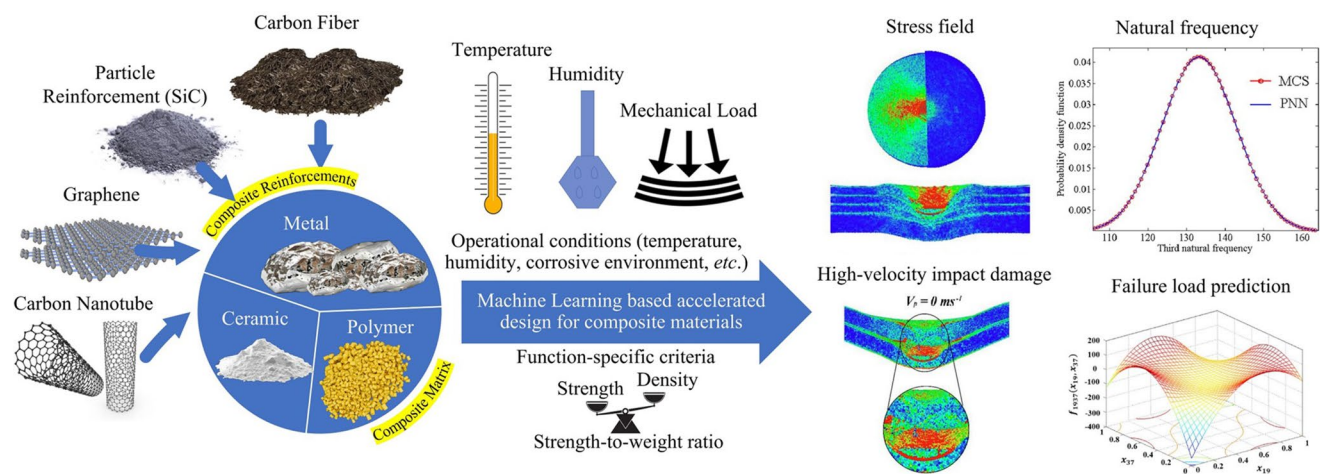


Fig. 4 Integrating data-driven methods for holistic mechanical and multi-physical characterization of composites. The performance of engineered composite materials depends on a variety of factors, including the concentration and spatial distribution of reinforcement agents, as well as variations in operating conditions such as temperature, humidity, and applied load. Computationally efficient machine learning models are required to accurately capture the impact of these parameters on performance and achieve function-specific, multi-crite-

ria optimization (e.g., high strength with low density). These models are essential for directing the design and analysis procedures, facilitating performance-driven material development and well-informed decision-making. The ML models can be exploited for different classes of composites, such as continuous and particulate fiber composites, as well as mechanical and a range of various other physical property characterization

the multimodal information from the FE simulation images of the UD composite. They carried out two-fold tasks with the framework, first prediction of the property, and second, capturing the spatial uncertainty in the microstructure of the UD composites. Apart from the prediction, deep learning is also employed to incorporate computer vision-based frameworks in the design and development of composite materials. For example, Bhaduri et al. [93] utilized an encoder-decoder framework for predicting the localization of stresses in UD composite by utilizing the image of the 2D representation of the fibres distribution. Gupta et al. [94, 95] utilized a generative-adversarial network (GAN) to predict the full-field stress distribution in the short-fibre polymer composites.

The studies reviewed in this section highlight the recent attempts to utilize the wide varieties of AI frameworks, ranging from classical ML algorithms to advanced computer vision deep learning models, concerning composites. This indicates the rising interest of the material research community in harnessing cutting-edge AI solutions in composite design.

4.3 AI-Driven Design of High Entropy Alloys

High entropy alloys (HEA) represent a novel class of materials which consists of multiple principal elements. Note that by strict definition, HEAs are also a special class of composites. Here, the presence of multiple elements in random concentration introduces configurational entropy, which results in several beneficial effects on the atomic level, such as the high-entropy effect, severe lattice distortion, and sluggish diffusion [96]. This novel class of material has shown promising potential in catering wide range of applications,

such as lightweight aerospace structures and engines [97], lightweight automotive components [98], nuclear materials capable of withstanding high-temperature instability and nuclear radiation damage [99], *etc.* The design of HEAs for the desired function-specific properties requires solving a combinatoric optimization problem, wherein the concentration of individual constituent elements has to be carefully decided. Moreover, the concentration of individual constituent elements also influences the microstructure of the end product. Hence, the optimal selection of the concentration of the constituent elements significantly affects the structure-property relationship of the HEAs. The computational capability of applied materials informatics can accelerate the function-specific alloy design (refer to Fig. 5).

To address such challenges, Shargh et al. [100] employed a deep learning model to classify the composition-specific phases (such as FCC, BCC, *etc.*) of the HEAs consisting of the elemental space of Ti, Al, Nb, Zr, Ni, V, and Fe. They used the large-scale dataset generated with the help of CALPHAD simulations for the construction and validation of the deep learning model. This study was further extended to deep learning-based refractory HEA discovery for finding the stable compositional space of Ti, Al, Nb, Zr, Ni, V, and Fe to ensure a single-phase BCC solid solution [101]. Rafiei et al. [102] presented a deep neural network-based framework to successfully predict the dislocation-driven stress fields and cross-interactions of dislocations. They used the dataset generated from discrete dislocation dynamics (DDD) simulations to carry out the investigation. The processing-structure-property relationship is a longstanding problem in the design and discovery of material systems for desired applications. To mitigate this challenge, Thoppil et al. [60] proposed a machine learning-based predictive framework

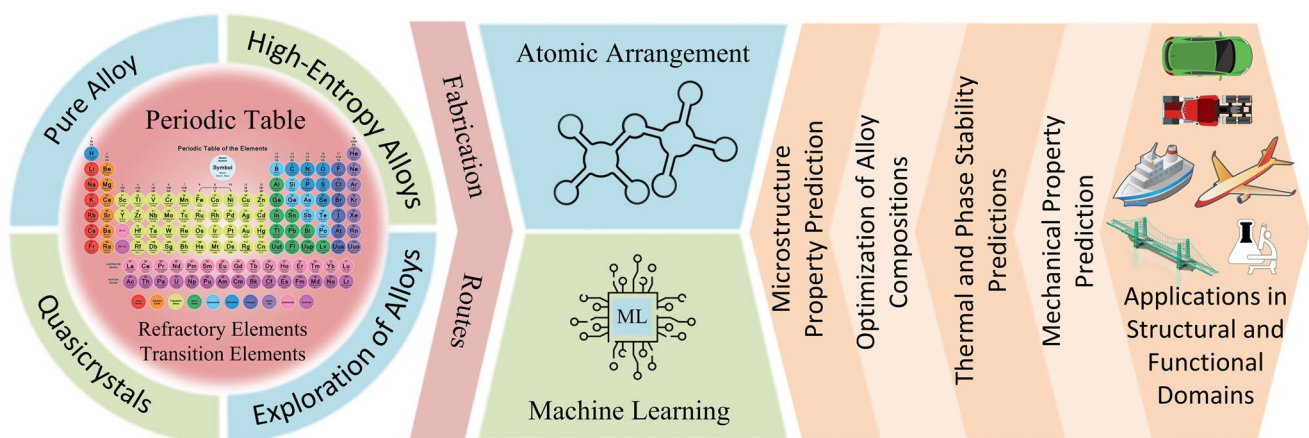


Fig. 5 AI/ML driven design of high entropy alloys for enhanced functional performance. Designing function-specific alloys using materials-informatics techniques has great potential because it allows for thorough characterization by accurately capturing changes in synthesis parameters and atomic configurations. Microstructures, elemental compositions, and thermodynamic properties must be carefully tuned

in order to create high-entropy alloys (HEAs) that are suited for particular applications. This process can be greatly accelerated by using a materials-informatics pipeline, which makes it easier to design and discover advanced alloys with multifunctional properties that are appropriate for use in the structural, electronics, energy, automotive, and aerospace industries

for accurately predicting the mechanical performance (fracture, yield) of complex concentrated alloys. They used processing conditions, compositional information (grain size and volume fraction), phase, density and stacking fault energy as the input features to map the process-structure-property linkages. Revi et al. [103] integrated the Materials Project repository, machine learning, and density functional theory (DFT) calculations to predict the elastic constants of high entropy alloys. To propose a rational and robust framework for alloy design, Mohanty et al. [51] utilized the least number of experimental results and applied the Bayesian approach to find the optimal composition of the alloys for maximizing the mechanical performance of alloys. Beausoleil et al. [104] demonstrated the application of materials-informatics concepts for accelerated design of high entropy alloys in nuclear applications. Zhang et al. [105] reported a binary classification model to predict the compositional space of FeCrNiCoMn HEAs with high yield strength. They used the uniaxial tensile simulation dataset generated by performing 300 molecular dynamics (MD) simulations. The transfer learning methodology can cut down the computational cost of training generalized deep learning models to a significant extent. In this regard, Jha et al. [106] proposed a pre-trained neural network to design the alloy systems based on the inherent chemical information associated with the chemical elements. Such a chemical domain-aware neural network presents a generalized approach to the scalable design of compositional materials. Wen et al. [107] proposed a data-driven surrogate model approach to optimally design $\text{Al}_x\text{Co}_y\text{Cr}_z\text{Cu}_u\text{Fe}_v\text{Ni}_w$ for enhanced hardness. The inverse design of materials enables the function-specific exploration of a vast parametric regime to obtain the optimal solution. Gupta et al. [108] proposed an end-to-end ML-driven pipeline for finding the optimal composition space of AlCoCrFeNi HEA for enhanced ballistic performance. The proposed computational framework is validated using observations obtained from MD simulations. A similar ML pipeline was extended later to design the AlCoCrFeNi HEA for a high stiffness-to-density ratio [109].

5 Artificial Intelligence Frameworks in Material Informatics

As discussed in the preceding section, the technological advances in computational prowess, in the past few years, have enabled the materials research community to develop AI-driven frameworks for accelerated design of functional materials. In this pursuit, the evolution of the AI framework has demonstrated great potential in exploring the uncharted territory of materials. This section presents an insightful understanding of different AI algorithms and their

evolution from machine learning to cutting-edge generative AI (GenAI) frameworks. The aim here is to supplement the discussions on AI-driven material discovery by contextualizing the AI algorithms rather than reproducing their mathematical foundations.

5.1 Machine Learning

Machine learning (ML) algorithms can be primarily categorized into *supervised* and *unsupervised* approaches. In the cases of supervised ML, the aim is to establish the relationship between dependent and independent variables with the help of labelled classes (in classification problems) or the continuous values of the dependent variables (in regression problems). In general, the supervised ML approach is guided by the available information in terms of input and target variables. Such ML models play a vital role in the development of the predictive framework, which is capable of predicting the target values for any random configuration of the independent variables. The supervised ML models can be further classified, as discussed below [110].

5.1.1 Parametric ML Models

The ML models which provide a closed-form solution as a mathematical equation are known as parametric models. Such models result in a mathematical relationship between the target property (Y) and input features (IF : x_1, x_2, x_3, x_4 , etc.) as:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \quad (1)$$

Here, $\beta_{(0-4..)}$ values denote the learned (through ML algorithm) weights of the input features, also known as coefficients. It is a general practice to use parametric ML models in cases where the correlation between the target property and input features is known. Basic models such as simple linear regression [111], multivariate linear regression [111], polynomial regression [112], and logistic regression [113] (for classification tasks) come under the category of parametric ML models. The parametric models are straightforward to develop and offer ease in interpretability, but these models mostly lack in capturing the complex relationships between the input features and the target response. The parametric ML models are mostly trained through gradient-based optimization approach (gradient descent optimization) to obtain the set of coefficients ($\beta_{(0-4..)}$) which yields a minimized loss function.

The parametric ML models are deployed in cases where simplicity and interpretability are preferred; these models have the inherent nature of explaining predictions based on the feature-dependent coefficients. The critical mathematical

understanding of the parametric ML models is presented in Table 2.

5.1.2 Non-Parametric ML Models

Unlike parametric ML models, the non-parametric ML models do not offer the closed-form solution (mathematical equation) of the relationship between the input features and the target response; rather, these models utilize probabilistic algorithms to efficiently capture the trend of the dataset. In terms of materials-informatics, these models demonstrate a promising potential in capturing the unknown, intricate physical phenomena of the materials. For example, a parametric ML model can establish the relationship between the alloy composition and wear rate by assigning specific weights to each compositional element, reflecting their importance. In contrast, a non-parametric model provides a probabilistic distribution of potential solutions rather than a single, fixed outcome. This flexibility makes non-parametric models more adaptable to changes in the data [114].

Some of the examples of non-parametric models are K-nearest neighbour (KNN) [115], support vector machine (SVM) [116], decision trees [117], Gaussian process regression (GPR) [118], *etc.* In the implementation of alloy composition-dependent wear rate, the plan of attack that KNN follows is that it finds the solution by averaging for the ‘*k*’ number of the neighbours in the vicinity of the available training data, whereas, decision trees (DT) work by recursively splitting the training data into subsets using the levels of the different features involved (alloy composition) and perform the averaging of the solutions to predict for the unseen data. The SVM employs the kernel trick to represent the dataset in higher higher-dimensional space for obtaining the flat hyperplane with the minimum errors possible, which makes the SVM model less prone to overfitting. The GPR model utilizes a Bayesian approach by establishing the

relationship between input features and target responses by providing the probability distribution over functions, which facilitates the evaluation of the uncertainty associated with the model predictions. Figure 6 provides illustrations of the aforementioned ML models. The training process of the non-parametric ML models involves hyperparameter tuning for the selection of the optimal hyperparameters to minimize the loss function. The critical applied insight on the discussed non-parametric ML models is presented in Table 3.

5.2 Deep Learning

The ML models offer major advantages in terms of sufficient generalization capability by utilizing a lower number of samples for training, provided the considered input features efficiently capture the complete envelope of the parametric range [119, 120]. In the context of materials science, where the availability of structured dataset is not always evident, the traditional ML models have the capability of developing predictive frameworks by utilizing sample space obtained from the carefully designed experiments/simulations [70]. Even though the use of ML models offers such advantages, they lack in efficiently capturing the complex relationship of high-dimensional feature space. In instances where a high-dimensional feature space is to be mapped with the material responses, or a feature space is available in different modalities such as image, signal, embeddings, *etc.*, it is essential to use complex AI frameworks such as deep learning (DL).

In this context, it may be noted that Deep Learning is a subset of Machine Learning. While Machine Learning uses algorithms like decision trees or SVMs and often requires manual feature extraction, Deep Learning uses multi-layered neural networks to automatically learn features from large amounts of data. Although it takes greater computational resources and training time, deep learning is excellent

Table 2 The mathematical understanding of the parametric ML models

Parametric ML model	Training philosophy	Mathematical Equation	Hyperparameters
Linear Regression [111]	Learns a linear mapping between input features and a continuous target variable by minimizing the sum of squared errors.	$\hat{y} = \beta_0 + \sum_{i=1}^n \beta_i x_i$ $\hat{y}$: predicted output β_i : model coefficients (weights) β_0 : model intercept (bias) x_i : input feature values	Regularization parameter (to prevent overfitting)
Polynomial Regression [112]	Optimization-based extension of linear regression; transforms features into polynomial terms to capture nonlinear relationships while still fitting a linear model on those transformed features.	$\hat{y} = \beta_0 + \beta_1 x + \dots + \beta_p x^p$ p : degree of the polynomial β_i : coefficients for each polynomial term x : input variable	polynomial degree (<i>p</i>), regularization parameter
Logistic Regression [113]	Discriminative learning; models the probability that the target belongs to a particular class using the logistic (sigmoid) function, and learns parameters by minimizing cross-entropy (log-loss).	$\hat{y} = \frac{1}{1 + e^{-\left(\beta_0 + \sum_{i=1}^n \beta_i x_i\right)}}$ $\hat{y}$: predicted probability β_i : learned weights x_i : input features	Regularization strength (<i>C</i> or λ),

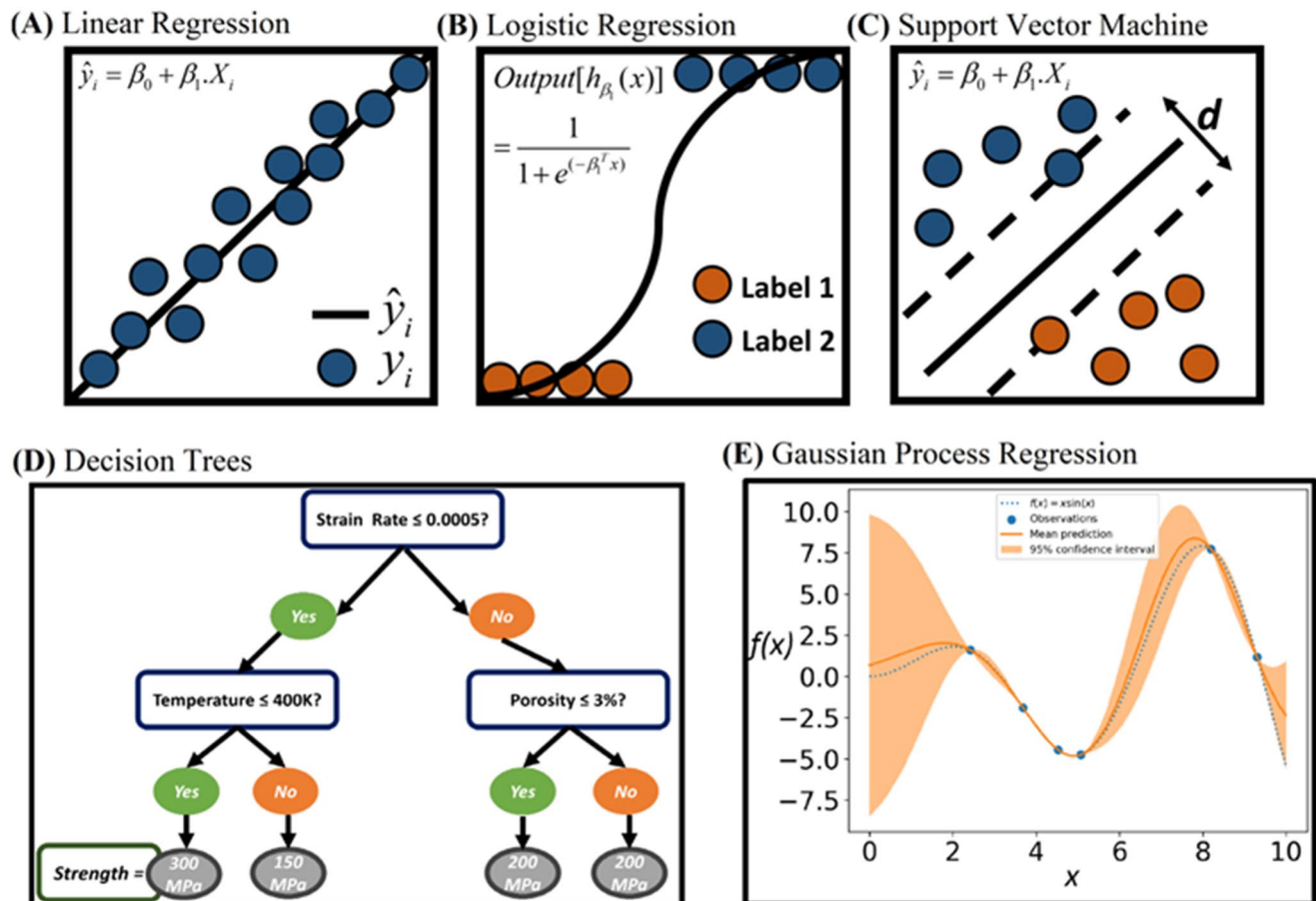


Fig. 6 Illustration of learning mechanisms in various machine learning models. **(A)** Linear regression aims to identify the best-fit line that models the relationship between input features and the target response. It produces a mathematical representation of this relationship in the form of a linear equation, allowing for straightforward prediction. **(B)** Logistic regression builds upon the approach used in linear regression but applies a logistic (sigmoid) function to the output. This transformation is appropriate for binary or multi-class classification tasks since it converts predictions to a probability scale. **(C)** The kernel-based support vector machine (SVM) technique is intended for both regression and classification. In order to guarantee a distinct division between classes, it aims to optimize the margin d between them. SVM improves

predictive generalization in regression by attempting to fit as many instances as possible within a given margin. **(D)** Decision trees create a tree-like structure by partitioning the feature space recursively and conditionally. Interpretable and hierarchical prediction-making is made possible by the decision rules that are represented by each internal node, which are based on feature values. **(E)** Another kernel-based technique is Gaussian process regression (GPR), which produces probabilistic predictions with corresponding uncertainty bounds as opposed to models that only produce one prediction. These bounds show the extent of possible error and provide information about each prediction's degree of confidence

at handling complex, unstructured data, such as text, audio, and images. Traditional machine learning techniques, on the other hand, are typically easier to understand and can perform well on smaller datasets. Simply put, all deep learning is machine learning, but not all machine learning involves deep learning.

5.2.1 Fully Connected Neural Network

DL models are powered by neural networks, consisting of fundamental building blocks that perform iterative feed-forward computations and backpropagation to optimize hyperparameters, enabling efficient learning. Neural networks

mimic the biological process of information transfer, with individual neurons playing a critical role in transmitting information. The basic version of DL is a fully connected neural network (FCNN), which is suitable for handling tabulated datasets for performing regression or classification tasks. The FCNN is a multilayered architecture of neural networks, consisting of one input layer, a series of hidden layers, and finally the output layer. In this series of layers, each neuron in the previous layer is fully connected with the neurons in the next layer. The calculation at the neuron level consists of a linear equation of the weighted sum of the input parameters and bias (similar to the linear regression), which is then passed on to the next layer after applying the

Table 3 The mathematical understanding of the discussed non-parametric ml models

Non-parametric ML model	Training philosophy	Mathematical Equation	Hyperparameters
k-nearest neighbor (KNN) [115]	Lazy learning (memory based), No learning, predicts by comparing distance of the test point with the training samples	$\hat{y}(x) = \frac{1}{k} \sum_{i \in N_k(x)}^n y_i$ $\hat{y}(x)$: predicted value for input x k : number of nearest neighbors $N_k(x)$: set of indices of the k nearest neighbors of x y_i : target values of those neighbors	No. of neighbors (k)
Decision Trees (DT) [117]	Recursive partitioning and region based predictions, suitable for both regression and classification tasks	$\hat{y}(x) = \sum_{m=1}^M c_m \mathbb{I}(x \in R_m)$ M : total number of terminal (leaf) regions, R_m : the m^{th} region of feature space, c_m : the predicted value (constant) for region R_m , $\mathbb{I}(\cdot)$: indicator function, equals 1 if condition is true, else 0.	maximum depth, minimum samples per split, minimum samples per leaf, maximum features, split criterion, splitter strategy (best or random)
Support vector machine (SVM) [116]	Actively learns by solving an optimization problem to find a maximum-margin hyperplane that separates classes (or fits data within an ε -tube for regression), suitable for both regression and classification tasks	<i>Hyperplane equation:</i> $f(x) = w^T x + b$, <i>Prediction rule:</i> $\hat{y} = \text{sign}(f(x))$ <i>Optimization objective:</i> $\min_{w, b, \xi_i} \frac{1}{2} w^2 + C \sum_{i=1}^m \xi_i$ <i>Constraints:</i> $y_i (w^T x_i + b) \geq 1 - \xi_i$, $\xi_i \geq 0$	Regularization parameter (C), kernel type (polynomial, radial basis function, sigmoid), epsilon (ε)
Gaussian Process Regression (GPR) [118]	Probabilistic Bayesian Learning, learns a distribution over possible functions that could explain the data	$f(x) \sim GP(m(x), k(x, x'))$ $m(x)$: mean function (often 0) $k(x, x')$: covariance (kernel) function	Kernel function ($k(x, x'; \theta)$), length scale (l), signal variance, noise level, mean function

activation function (such as sigmoid, tanh, rectified linear unit (ReLU), leaky ReLU, *etc.*). The application of the activation function is essential to ascertain the distribution of the weighted sum response between certain intervals, such as -1 to $+1$ in the case of the sigmoid, 0 to 1 in the case of ReLU, *etc.*

The complete process of weighted sum and implementation of the activation function is known as the feed-forward mechanism of the information transfer from the input layer to the output layer (refer to Fig. 7(A)). The weighted and transformed output of the feed-forward mechanism (at the output layer) is compared with the actual output to calculate the loss (prediction error). Now, the next process is to optimize the weights and bias to minimize the loss, which happens through the following mechanism of backpropagation (refer to Fig. 7(B)). The output obtained from the feed-forward mechanism is back-propagated to update the weights and bias. This combined process of one feed-forward and one back propagation is termed one epoch; performing the iterative repetition of these epochs leads to minimization in loss and simultaneous enhancement in the predictive accuracy [119]. Such neurons are a unit block of dense neural networks, where each neuron in the previous layer

is connected to the neurons in the following layer, hence named as a Fully Connected Neural Network (FCNN) (refer to Fig. 8(A)).

5.2.2 Convolution Neural Network

To capture the relevant information from the images or spatial dataset, the FCNN is updated with two important layers before the input layer of the FCNN, namely the convolution layer and the pooling layer. The process of convolution involves applying the kernel (also known as a filter) to the input image, which is represented as a matrix of pixel values. For instance, if the provided image has a matrix size of 32×32 (referring to the pixel size), in a convolution operation, a kernel (e.g., 3×3 or 5×5 matrix) slides over the image and performs element-wise multiplication followed by summation. The prime objective of this operation is to capture the spatial information from the given image, such as patterns, edges, *etc.*, which is known as a feature map. The captured spatial features are then applied with the *ReLU* activation function, which introduces non-linearity to the model, enabling it to learn complex patterns. The convolution operation is followed by a pooling operation, which is

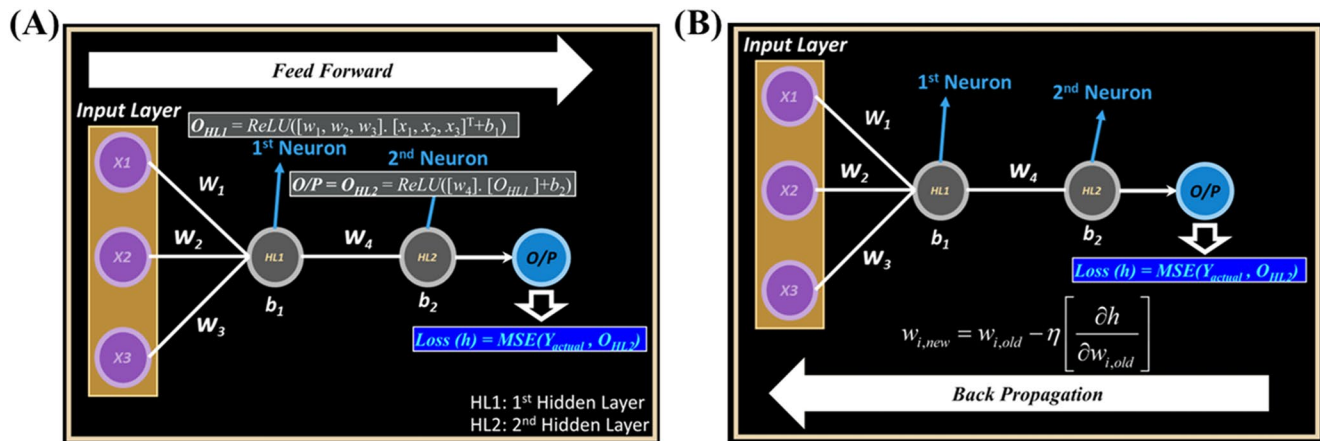


Fig. 7 Learning process at the neuron level of a neural network. (A) *Feedforward mechanism*: a weighted sum of input parameters and bias, followed by application of an activation function to capture the nonlinear relationship between input features and the target response. (B) *Back propagation*: optimally updating the weights and bias for minimizing the prediction loss. In this context, the gradient of loss

with respect to an individual weight/bias is evaluated, which is then used to find the global minima (gradient ≈ 0) by deploying gradient descent optimization (or any other optimizer). The iterative feedforward and back propagation is performed to find the optimal weights and bias representing the learned model

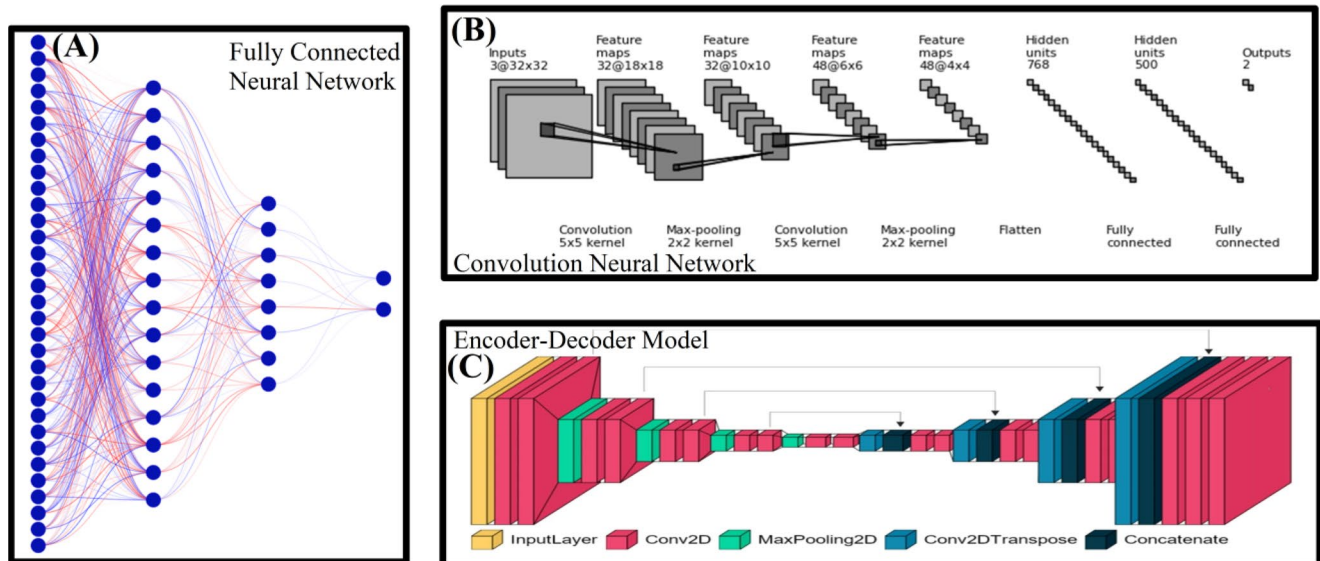


Fig. 8 Illustration of learning mechanisms in various deep learning models. (A) The weighted sums and activation functions are used to connect each neuron in the multiple-layer fully connected neural network (FCNN) [121]. Loss is calculated by comparing the model's output with the actual value; backpropagation through iterative epochs minimizes this loss, gradually increasing prediction accuracy. (B) In order to extract spatial features from images, the convolutional neural network (CNN) [122] builds upon the FCNN by adding convolution and pooling layers prior to the input layer. After applying a kernel to the input image to extract patterns, the convolution layer applies the ReLU activation to account for non-linearity. By decreasing the feature map's spatial dimensions, pooling increases productivity and lessens overfitting. CNNs are perfect for tasks like image classifica-

tion and object detection because they use stacked convolution and pooling layers followed by a fully connected network for prediction. (C) The convolution and pooling layers of the CNN architecture are used in sophisticated frameworks such as the encoder-decoder model [122]. The decoder uses up-sampling and transposed convolutions to reconstruct the image, with skip connections improving reconstruction accuracy, while the encoder extracts low-level features and compresses the image into a latent space. This architecture is essential for tasks like image segmentation and image-to-image translation (e.g., Pix2Pix GAN), where a discriminator uses adversarial training to guarantee high-quality outputs while a generator learns to map one image domain to another

used to reduce the spatial size of the obtained feature map. This is typically performed through max pooling or average pooling, where the maximum or average value of a region of the feature map is selected, respectively. Pooling helps reduce the computational load and controls overfitting by abstracting the high-level features. For example, pooling could reduce a 32×32 feature map to 16×16 , significantly reducing the number of parameters in the model.

Most often, the stack of multiple “convolution+pooling” layers is used to extract the complex features from the input image. Post-stacked convolution and pooling layers, the stacking of a deep neural network (fully connected neural network) is used to perform the same feed-forward and back-propagation operations [120]. This architecture of the neural network is called Convolution Neural Network (CNN) (refer to Fig. 8(B)), which is instrumental in computer vision tasks such as image classification and object detection. Advanced CNN architectures, including AlexNet [123], VGG [124], and ResNet [125], have pushed the boundaries of these tasks by incorporating deeper layers, residual connections, and novel approaches like batch normalization and dropout to increase training speed, stability, and generalization performance.

5.2.3 Encoder-Decoder Models for Handling Image Datasets

The convolution and pooling concepts of CNN architecture (explained in subsection 5.2.2) are directly applied to some of the advanced frameworks such as encoder-decoder models [126], where the encoder performs a series of combined convolution and pooling (down-sampling) operations to extract the low-level features such as edges, texture, corner, *etc.* (refer to Fig. 8(C)). The encoder part of the model results in a compressed latent space of the image (a bottleneck) - for instance, an encoder would compress a 256×256 image into a 16×16 feature map. The compressed representation captures high-level features of the image. In the decoder, transposed convolutions (also termed deconvolution) and up-sampling are performed to reconstruct the image back to its original or target size. In this process (of the decoder), skip connections are used to preserve spatial information by passing features from the earlier convolution layer directly to later stages. The skip connections help in improving the reconstruction accuracy of the image.

Such an encoder-decoder architecture is useful in performing image segmentation, and image-to-image translation (pix2pix generative adversarial network, i.e. pix2pix GAN), where the model learns to map one image domain to another [127–129]. Pix2Pix GAN works by training a generator to convert input images (e.g., sketches) into realistic outputs (e.g., photos), while a discriminator learns

to distinguish real from generated images. Both networks improve together through adversarial training, using paired image datasets to learn accurate, pixel-level translations between domains. Here, the generator is built using an encoder-decoder architecture, while the discriminator evaluates whether the output is realistic, enforcing high-quality generation. Additionally, similar encoder-decoder models in machine translation are used for sequence-to-sequence tasks, learning to map input sequences (e.g., text in one language) to output sequences (e.g., translated text).

5.2.4 Natural Language Processing (NLP)

The utilization of deep learning frameworks in natural language processing (NLP) requires specialized architecture, where previous information is stored in terms of memory units. In the early stages, a recurrent neural network (RNN) was utilized to capture the language sequences and predict the next token (word) [130]. This architecture utilizes hidden states as the memory unit, making them capable of handling short-term dependencies, such as completing sentences. However, they struggle with long-range dependencies and are prone to the vanishing gradient problem during back-propagation. The gradients (rate of change in loss function with respect to the change in the hyperparameter) help in the weights updation process, which helps in reaching sufficient accuracy in the predictions. With the vanishing gradient (gradient becomes very small) problem, the solutions are stuck at the local minima without updating the weights any further, which hinders acquiring high predictive accuracy. To address these challenges, Gated Recurrent Unit (GRU) [131], and Long-Short Term Memory (LSTM) [132] architectures were introduced, providing better memory management and mitigating the vanishing gradient problem. LSTMs and GRUs still have drawbacks despite these improvements, including ineffective parallelization of training and difficulties in capturing the contextual meaning of words - a word's meaning might vary based on its context.

With the recent advent of transformer architecture, these challenges were effectively mitigated by using the attention layers [133]. Initially, the Transformer architecture was proposed to perform the machine translation task. This architecture provides the positional encoding to the word sequence and utilizes the encoder-decoder framework, where the encoder consists of a multi-head self-attention layer, feed-forward neural network, and addition & normalization layers, whereas the decoder architecture consists of masked multi-head self-attention layers, multi-head cross attention layers, a feed-forward neural network, and addition & normalization layers. The output layer of the transformer architecture consists of the SoftMax function to evaluate the probability of the next token (prediction) in the

sequence [134]. The transformer architecture considers the positional encoding of the words in the sequence, to capture the sequential order of the tokens (i.e. near proposition, or near verb, *etc.*) - this helps in providing the first hint of the contextual meaning of the tokens used in the sequence.

The self-attention layers in the encoder also allow for capturing the contextual meaning of the used word by finding the relation of each token with the other tokens available in the sequence. The attention layers apply the query, key and value matrices to the positionally encoded tokens to add a higher number of tuneable parameters, which are then processed in the feed-forward neural network of the encoder. The output of the feed-forward neural network is then added to the input of the layer (by using the residual connection) and normalized. The normalized output of the encoder is fed to the masked multi-head self-attention layers of the decoder, enabling the decoder to perform autoregressive prediction of the next tokens. The masking in the self-attention layer of the transformer decoder is crucial for ensuring that the model does not have access to future tokens when making predictions. This is essential for autoregressive tasks, where the model generates sequences token by token, with

each token being predicted based on the previous tokens. Next, the output of the masked multi-head self-attention layer is compared with the output of the self-attention in the encoder by using the cross-attention layers. Again, the feed-forward neural network is used in the decoder to process the information obtained from cross-attention layers. The attention layers used in the architecture are the core of the transformer, which provides a large number of tunable parameters to enhance the accuracy of the language model. This advanced architecture has significantly progressed language processing tasks. Unlike the RNN architecture, which processes one token at a time, making it inefficient for handling long-range dependencies, the Transformer processes the entire sequence of words in parallel using attention layers. This parallel processing not only improves the model's ability to capture dependencies but also speeds up training compared to the LSTM or RNN frameworks.

The two different variants of Transformer models are illustrated in Fig. 9. Figure 9(A) highlights the utility of the encoder block of the transformer model, which encodes the materials data into numerical embeddings along with positional encoding. The numerically encoded information is

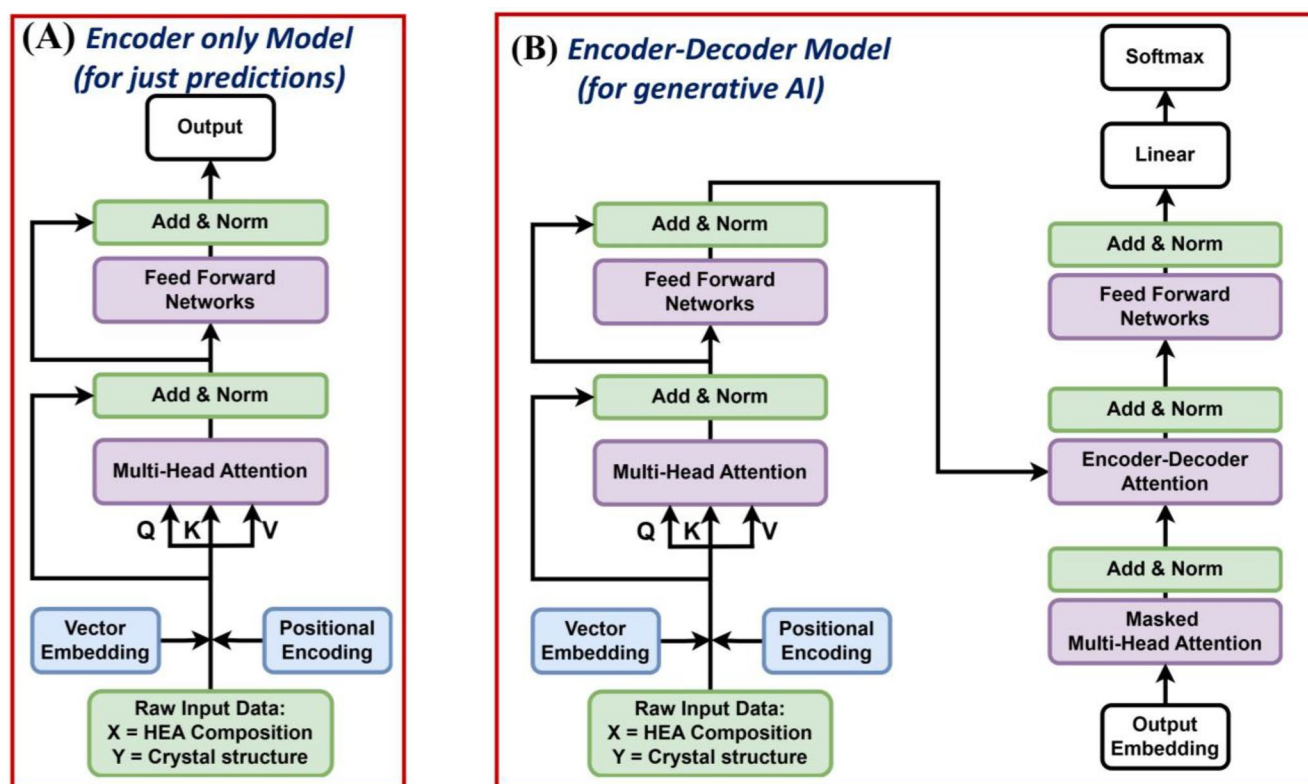


Fig. 9 Variants of transformer models. (A) The encoder-only models utilize the encoder block of the transformer model, which encodes material data into numerical embeddings with positional encoding, followed by the calculation of attention scores in the multi-head attention block. While encoder-only models excel in property prediction, they lack the generative capabilities of full transformer models. (B) Encoder-decoder (full transformer) is used to introduce generative

capabilities, wherein the decoder generates a suitable response based on the natural language query processed by a properly finetuned encoder. Both the variants, encoder-only and encoder-decoder utilise the attention mechanism for understanding the key connections within the attributes used in the calculation of attention scores in the attention block - Q , K , and V representing query, key and value matrices, respectively

then applied with query, key and value matrices to calculate the attention scores in the multi-head attention block. The calculated attention scores are then fed to the feed-forward neural network. The encoder-only model is useful in making the property prediction, but lacks the generative capability of the transformer model. The recently reported materials domain-aware finetuned models, such as MaterialsBERT [135], MatSciBERT [136] and MatBERT [137] utilized the encoder-only framework to train the encoder over the materials dataset. The generative capability of the Transformer architecture utilizes a combined architecture of encoder-decoder, wherein the encoder is trained with the materials science corpus - for instance, the published literature on the high entropy alloy, and the decoder is capable of predicting the response as per the user prompt (in natural language) (refer to Fig. 9(B)).

In summary, Natural Language Processing (NLP) is a broad field within artificial intelligence that focuses on enabling machines to understand, interpret, and generate human language. It includes tasks like text classification, translation, sentiment analysis, and speech recognition, using rule-based, statistical, or machine learning approaches. Large Language Models (LLMs), such as GPT or BERT, are a specific type of deep learning model within NLP. Trained on massive text datasets using transformer architectures, LLMs excel at generating and understanding natural language. In essence, NLP is the overall discipline, while LLMs are advanced tools developed within it to perform language-related tasks more effectively. In sect. 6, we further highlight the evolving trends of exploiting LLMs in materials discovery.

5.3 AI-Driven Pipeline for Materials Design

In recent years, there has been a remarkable surge in the use of artificial intelligence (AI) frameworks for materials design and discovery. The regime of materials informatics has been completely transformed by the recent advances in computational capability, moving from the analysis of traditional tabulated datasets to the integration of various data modalities like text, images, and other complex forms. This subsection presents the AI-driven pipeline designed to handle and interpret different data modalities in the materials science domain. Historically, the primary input for materials informatics workflows has been tabulated datasets, which are typically sourced from published sources, computer simulations, or laboratory experiments. To choose the relevant features from the raw dataset that significantly affect the problem statement, feature engineering is the next step. The feature engineering involves the common practices such as correlation analysis for selecting the relevant features and principal component analysis (PCA) for feature

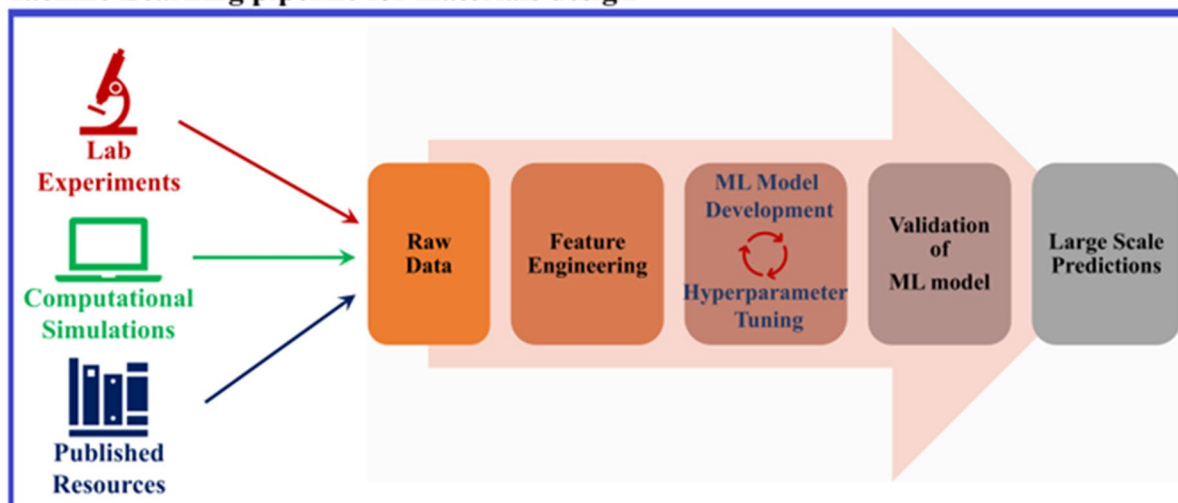
extraction (reducing the number of features without losing information). The pre-processed dataset is used to train the optimal ML model by enforcing hyperparameter tuning, and finally validated with the out-of-fold data (separately kept Test data). Once the predictive performance of the trained ML model is rigorously evaluated and confirmed, it can be deployed for large-scale predictions in downstream materials design and discovery applications. The conventional ML pipeline for handling the tabulated dataset is presented in Fig. 10(A).

The published resources in materials science often contain diverse data formats, including alphanumeric strings (such as compositional and structural formulas), images, and textual descriptions that capture process–structure–property (PSP) relationships. The conventional ML pipelines, however, are typically limited in their ability to process such multimodal information effectively. To address this challenge, the preprocessing stage of the AI-driven pipeline incorporates vectorization, the transformation of heterogeneous data types into a unified numerical feature space, thereby enabling the development of predictive models capable of learning from multiple data modalities. The featurization schemes facilitated by composition-based feature vectors (CBFV) [51], or MatMiner [27] utilizes the constituent elements' specific chemical descriptors to represent the global fingerprint of the compositional materials. Likewise, the molecular fingerprint of the organic compounds (SMILES) is produced by Molecular Access System (MACCS) keys [138], which results in high-dimensional binary string representing the presence or absence of the functional groups in the compound. To numerically vectorize the materials' specific images (microstructural representations), the conventional image handling schemes, such as feature extraction through convolution operations, are performed [94]. The materials' textual corpora are vectorized by generating the relevant embeddings from the materials domain-aware BERT models, such as MaterialsBERT [135], MatSciBERT [136], etc. The numerical vectorization scheme is generally integrated with the deep fully connected neural networks (FCNN) for effectively learning the correlations among the high-dimensional feature space and the target property. The deep learning pipeline for the multi-modal information processing within the materials science domain is presented in Fig. 10(B).

5.4 Critical Remarks on Choosing the Right AI Model

The application of machine learning (ML) in materials science and discovery has significantly accelerated the identification, optimization, and design of materials with desired properties [139, 140], as discussed in the literature review

(A) Machine Learning pipeline for materials design



(B) Deep Learning pipeline for materials design

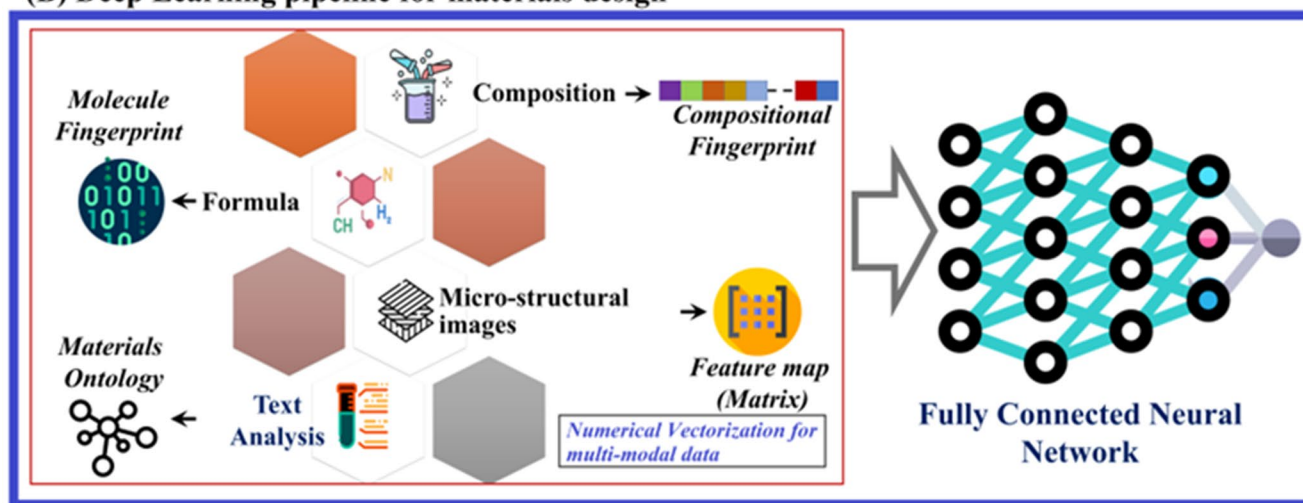


Fig. 10 AI-driven pipeline for materials design. **(A)** Machine learning for handling the conventional tabulated dataset (consists of input features and the target responses), wherein the dataset ingested from various sources is processed for feature selection/extraction; the processed dataset is used to train an efficient ml model by enforcing hyperparameter tuning and rigorous testing through unseen data. Finally, the developed ml model is deployed for making large-scale predictions.

(B) To handle the multi-modal information relevant to materials science, such as compositional and molecular formula, microstructural configurations, process-structure-property relations, the information-specific vectorization scheme is implemented to generate the unique embeddings/feature maps, which are then integrated with the FCNN for learning the relationship between high-dimensional feature representation and the target response

presented in Sect. 4. However, the effectiveness of ML models depends on choosing the right approach for the problem at hand, considering data availability, model interpretability, and computational complexity. A critical comparison of various ML paradigms, supervised, unsupervised, parametric, non-parametric, and deep learning, highlights their complementary strengths and limitations.

The most popular method in materials research is supervised learning, especially for predictive tasks like determining bandgap, thermal conductivity, or elastic modulus from compositional or structural features [141]. When trained on

labelled datasets derived from experiments or simulations (e.g., DFT), algorithms such as support vector machines, random forests, and fully connected neural networks (FCNNs) have shown high accuracy [106, 142]. Nevertheless, when trained on limited or skewed datasets, these data-intensive models may have poor generalization. Furthermore, unless they are supplemented with methods like feature attribution or symbolic regression, their predictions might not be physically interpretable [143].

Unsupervised learning, on the other hand, reduces dimensionality for visualization, finds hidden patterns in

spectroscopy or diffraction data, and clusters similar materials without the need for labeled data [144, 145]. Relationships in complex, high-dimensional materials datasets have been successfully revealed by methods such as principal component analysis (PCA), t-SNE, and k-means clustering. Nevertheless, these models provide little direction for design or optimization, and meaningful interpretation of the results usually necessitates domain knowledge.

For quick and easy-to-understand estimations in systems with relatively simple variable relationships, parametric models, like linear regression, assume a particular functional form. They are frequently employed to confirm mechanistic hypotheses or in early materials screening. They fall short, though, in describing the nonlinear behaviors that are commonly seen in actual material phenomena. Kernel techniques and Gaussian processes are examples of non-parametric models that do not assume a set form and can readily adjust to complex data. Though they are usually more computationally demanding and less interpretable, they offer superior accuracy in modeling nonlinear relationships, such as capturing phase stability or predicting fatigue behavior [146, 147].

In materials science, deep learning models are becoming more and more popular, especially when dealing with unstructured or image-based data. For the analysis of microstructural images from SEM, TEM, or XRD, Convolutional Neural Networks (CNNs) are very efficient, allowing for automated feature extraction and classification [148]. They are frequently referred to as “black boxes” and necessitate sizable, labeled datasets. On small datasets, Fully Connected Neural Networks (FCNNs) run the risk of overfitting, but they work well with structured numerical inputs [149]. Inverse design, where the objective is to create structures from target properties or reconstruct material configurations, is being investigated using more sophisticated architectures such as encoder-decoder models and pix2pix GANs [150]. Despite their exciting potential, these generative models have issues with data hunger, training instability, and physical interpretability.

Through the conversion of enormous volumes of unstructured scientific text into structured, actionable data, NLP is increasingly contributing to the acceleration of materials innovation. Literature mining is a crucial application in which NLP algorithms retrieve names of materials, synthesis techniques, characteristics, and performance indicators from patents and journal articles. This makes it possible to build databases of machine-readable materials that facilitate data-driven modeling and discovery. By examining publication patterns, NLP also supports trend analysis, which helps researchers find new materials or technologies. Complex scientific language can be understood by sophisticated models like BERT and GPT, which makes it easier to

generate hypotheses, automate literature reviews, and even predict new material-property relationships. Additionally, by matching functional targets with materials descriptors from text, NLP can facilitate inverse design. By unlocking latent knowledge embedded in publications and patents, NLP significantly reduces the time and effort required to explore new material spaces, making it a powerful tool for innovation.

In conclusion, no single model is always the best, even though machine learning (ML) provides effective means for materials discovery. Model selection must be in line with the nature of the issue, the availability of data, and the requirement for interpretability in order to be deployed effectively. Hybrid approaches that combine several machine learning models with domain expertise and physics-based modeling are the way of the future for materials innovation because they allow for precise prediction as well as scientific understanding.

6 Integrating Material Domain Knowledge with Contemporary AI Developments

6.1 Large Language Model (LLM) for Materials Knowledge Extraction

The exceptional capabilities of Transformer architecture have revolutionized natural language processing, and its application in materials exploration is also making significant progress, highlighting several interesting developments in the field. The potential of exploring the scientific literature by querying the relevant materials-specific information compels utilizing the language processing capability of the Transformer architecture. In this regard, a few attempts have been made to develop materials domain knowledge-aware transformer architectures. For instance, Shetty et al. [135] developed MaterialsBERT by finetuning the PubMedBERT [151] with the large polymer corpora compiled by using a large number of abstracts of the published papers. Gupta et al. [136] finetuned SciBERT [152] with the corpus of the published literature on families of materials such as inorganic glasses, metallic glasses, alloys, etc. and developed MatSciBERT. Wei et al. [137] proposed a transformer architecture (MatBERT) to construct, classify and property prediction for the compositional crystals. The probabilistic generation of crystals was validated with the DFT simulations. Huang et al. [153] developed MatInFormer, a generalized finetuned transformer architecture for materials property prediction. These few attempts focused on developing material domain knowledge-aware transformer models, which required rigorous finetuning. Finetuning the transformer models over a large corpus can be a computationally

expensive task; hence, some of the research groups exploited the sound generalization capabilities of pre-trained BERT models and large language models (LLM) (such as the llama series by Meta) for performing the materials-specific downstream tasks. Such as Trewartha et al. [154] compared the performance of Bi-LSTM, SciBERT, and MatBERT for the named entity recognition task. They reported the superior performance of MatBERT, due to its general awareness in the materials domain. Choudhary [155] devised an LLM (generative pre-trained transformer (GPT)) driven approach for forward and inverse materials design. Prasad et al. [156] utilized zero-shot prompting over LLM models such as Open AI GPT, and Gemini Pro to extract the materials-specific information from the published literature - this work leverages the general contextual understanding of the LLM models without finetuning them with the materials corpus. Few-shot prompting/Zero-shot prompting significantly reduces the computational burden of finetuning LLMs and still works significantly better in terms of knowledge extraction. The attention mechanism of the transformer model is used for ascertaining the context of the sentence - Wang et al. [157] utilized this capability of attention layers to find the best compositions of elements for different materials property prediction. They used the attention layers to find out which elements are highlighted (high attention scores)

the most while predicting the functional property-dependent composition of the materials. Chandrasekhar et al. [158] utilized the llama 2 7B parameters LLM model to develop the question-and-answer model by enforcing retrieval augment generation for additive manufacturing.

Figure 11 demonstrates the downstream deployment of MaterialsBERT [135] in “Named Entity Recognition (NER)” for automated extraction of the polymer-specific information, such as name, concentration, property-value pair, etc. The NER pipeline utilized MaterialsBERT (materials domain-aware BERT) for generating the text embeddings, which further helped in identifying the material-specific entities (such as the material’s name, concentration, property-value pair). Such automated extraction of the relevant information presents a paradigm-shifting revolution in the domain of materials research, which enables the strategic utilization of published research for accelerated review and discovery of materials.

In summary, this section demonstrates the successful utilization of different formats of large language models to process the materials-specific published literature. Such implementation of NLP techniques in the materials science domain shows great potential in leveraging the published literature for strategic extraction of critical information such as process, material, structure, property, etc. [159]. This

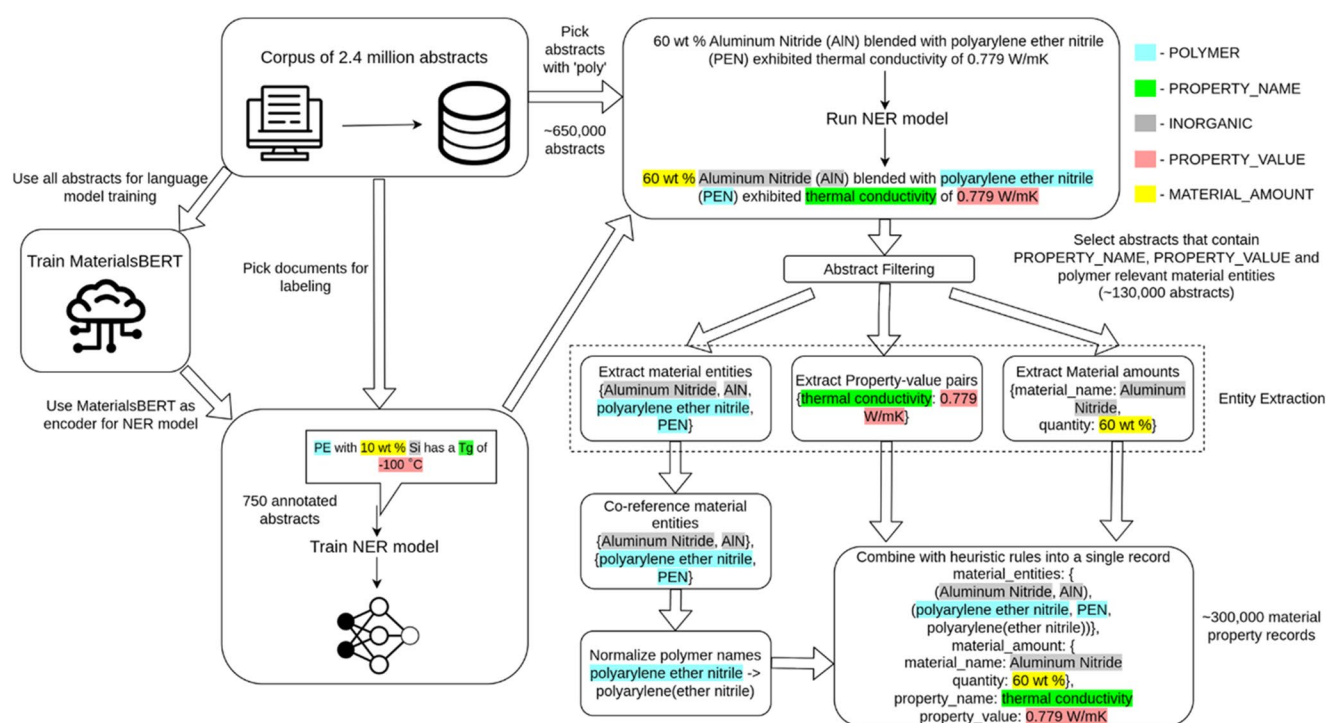


Fig. 11 MaterialsBERT for literature mining. BERT-based language models can be fine-tuned on domain-specific corpora within the materials science field. The resulting model, MaterialsBERT, can then be applied to named entity recognition (NER) tasks to extract materials-specific information such as material names, compositions, and prop-

erty-value pairs (Reproduced from Shetty et al. [135]). Deploying such natural language processing (NLP) models demonstrates significant potential in addressing a key challenge in data-driven research, such as the scarcity of high-quality, domain-specific datasets in materials science

could not only solve the problems of the non-availability of the relevant dataset but also devise a novel materials-informatics pipeline which allows the user to query the materials-specific information in natural language. The LLM-based material informatics pipeline could be used for various downstream tasks such as named entity recognition (NER), retrieval augmented generation (RAG), etc.

The deployment of LLMs in a materials-informatics regime facilitates the ingestion of materials domain knowledge in terms of processes, structure, and property. The long-standing difficulty in modeling PSP linkages originates from the inherent complexity of multimodal data processing. Such challenges are effectively solved by the materials domain-aware LLMs. Table 4 lists some of the recently developed LLMs or other variants of transformer architecture aimed at processing the natural language query associated with the materials terminologies.

6.2 Generative Artificial Intelligence in Materials Design and Innovation

Generative Artificial Intelligence (GenAI) is emerging as a transformative force in materials design and innovation, enabling the creation of novel materials by learning patterns from existing data and generating new candidates with desired properties. Unlike traditional predictive models, which forecast properties from known materials, GenAI inverts the design process, proposing new structures or

compositions tailored to specific functional requirements [106, 139].

Techniques such as Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Transformer-based models (e.g., GPT-like architectures) are being used to explore vast materials design spaces. For example, VAEs can generate new crystal structures by sampling from latent chemical spaces [165], while GANs (like pix2pix) are used for microstructure generation conditioned on target properties [166]. Transformer models are also being applied to generate polymer sequences, molecules, or synthesis pathways in text-like formats [167].

One of the key advantages of GenAI is its ability to propose candidates that may not exist in current databases, overcoming human design bias and expanding the search beyond well-explored material families [168]. These models can also be trained on multi-modal data, combining text, images, and numerical descriptors, offering a richer understanding of materials behaviour and enabling inverse design with unprecedented flexibility [143]. However, several challenges remain to be addressed. GenAI models often require large, high-quality datasets and may generate physically unrealistic or chemically invalid candidates without appropriate constraints [16]. As a result, there is a growing interest in physics-informed generative models, which embed domain knowledge and conservation laws into the learning process to improve trustworthiness and relevance [169]. Additionally, the interpretability of GenAI outputs

Table 4 List of materials domain-aware LLMs and their use cases

LLM Type (by Input Representation)	Example Model	Architecture	Primary data format	Application/Use Case
Process-based/Manufacturing LLMs	AMGPT [158] Chat-IMSHT [160] Intelligent Manufacturing Support [161]	RAG (LLM+retriever)	Process documentation, Q&A text	Manufacturing workflow Q&A
Composition-based	Crystal Composition Transformer [137]	Transformer-based Blank-Filling Language Model (BLMM)	Tokenized chemical composition strings	Generative materials composition design, doping and substitution (“tinkering”), and property-guided discovery of new compounds
Formula-based LLMs	SMILES-Transformer [162] Chemformer [163]	Encoder–decoder Finetuned BART	SMILES molecular sequences SMILES molecular sequences	Molecular property prediction inverse design
Structure-based	MatInFormer [153] Molecular Graph Transformer (MGT) [164]	RoBERTa-based Transformer Encoder Hybrid Graph–Transformer architecture with multi-head attention	Tokenized crystalline information combining space group tokens, informatics tokens (e.g., topology, porosity, lattice parameters), and formula tokens Graph-structured atomic data	Interpretable crystal property prediction, lattice parameter learning, and structure-aware materials representation without explicit atomic coordinates Predicting material and molecular properties
Text/Knowledge-based	MaterialsBERT [135] MatSciBERT [136] MatBERT [154]	Transformer’s Encoder-only (BERT-type)	Scientific text corpora	Named Entity Recognition, knowledge extraction

is a concern. Generated candidates must be validated not only computationally (e.g., through DFT or phase-field simulations) but also experimentally, which can be resource-intensive. Integration with active learning and closed-loop experimentation frameworks is being explored to guide experimental synthesis more efficiently [170].

In summary, generative AI represents a paradigm shift in materials science, enabling autonomous creativity in material design. When integrated with domain expertise, simulation tools, and experimental validation, GenAI has the potential to drastically shorten development cycles, uncover novel functionalities, and accelerate breakthroughs for materials innovation across sectors such as energy, electronics, aerospace, and healthcare.

6.3 Prospect of Agentic AI in Materials Design

Agentic AI is emerging as a transformative paradigm in materials science research, enabling autonomous, goal-directed reasoning across complex design, simulation, and experimentation workflows [171]. Unlike traditional machine learning or scripted automation, agentic AI systems can independently plan multi-step tasks, interact with diverse tools (e.g., simulation engines, databases, literature), and adapt strategies based on outcomes, mirroring aspects of scientific reasoning. This approach is particularly promising for tackling inverse materials design problems, such as discovering architectures that exhibit targeted multifunctional properties, where vast search spaces and multi-physics interactions challenge conventional methods.

Agentic AI works by acting as a goal-driven, autonomous problem solver that can reason, plan, and make decisions across multiple steps and tools to achieve a scientific objective. In materials science, this means that instead of simply running predefined tasks or analyzing datasets, agentic AI can take a high-level research goal, such as “design a lightweight, vibration-damping lattice metamaterial”, and break it down into sub-tasks like reviewing relevant literature, querying materials databases, running simulations, evaluating results, and proposing next steps. It can interact with software tools (e.g., CAD, FEA, DFT), use APIs to access experimental data or control robotic labs, and even revise its approach based on new findings. This makes it highly useful in materials research, where the design space is often vast and multidisciplinary, combining mechanics, chemistry, physics, and manufacturing constraints. Agentic AI can explore more design alternatives faster, uncover non-intuitive solutions, and streamline the path from concept to prototype. As materials challenges become more complex, such as achieving multi-functionality, sustainability, or adaptive behavior, agentic AI offers a powerful way to accelerate

innovation by connecting data, models, and experiments into a unified, intelligent workflow [172–174].

A parallel and complementary trend is the development of Autonomous Materials Laboratories (AMLs), which integrate AI-driven decision-making with robotic experimentation and real-time data acquisition [175, 176]. While AMLs excel in accelerating hypothesis testing and data generation, agentic AI can orchestrate broader research objectives, coordinating AMLs, simulations, and theoretical models. Looking ahead, the integration of agentic AI with AMLs and digital twins is poised to define the next generation of materials research, shifting from automated discovery to intelligent, end-to-end scientific exploration.

7 Evolving Trends, Scope of Improvement, and Future Research Trajectories

The evolution and adoption landscape of machine learning in the domain of materials science is presented in Fig. 12 based on the data analytics obtained from the Scopus database. A broad overview of publication statistics is shown in Fig. 12(A, B). A closer analysis of articles reveals that pre-2000 most of the articles demonstrated the use of bio-inspired algorithms such as genetic algorithm, artificial neural network for optimal design of materials (due to the limitations in computational capabilities), post 2000 the use of classical machine learning such as Polynomial Regression, Multi-Layer Perceptron, Decision Trees, Support Vector Machines etc. is prominently seen in materials research. It is only after the year 2010 that a rapid rise in publications is seen, which employed deep learning frameworks. Moving forward, this evolution has culminated in the era of generative AI in the domain of materials research. Figure 12(C) presents a detailed distribution of the publication statistics on different subfields of machine learning and artificial intelligence concerning materials design and discovery. The information is retrieved from Scopus by using the combination of keywords “Materials AND Machine Learning OR Deep Learning OR Large Language Models” resulted in ≈ 15000 Scopus-indexed research articles. A closer analysis of published literature revealed that pre-2000, the use of AI frameworks in materials science was scarce, which drastically increased post-2010. The presented analysis utilised the Scopus extracted abstracts with a look-out algorithm for searching a series of keywords associated with machine learning, deep learning, and large language models. The published literature demonstrates that post 2010, the machine learning and deep learning frameworks have taken a prominent place in materials design and discovery. Post 2020, the use of generative AI frameworks such as generative adversarial networks has seen a rise in implementation.

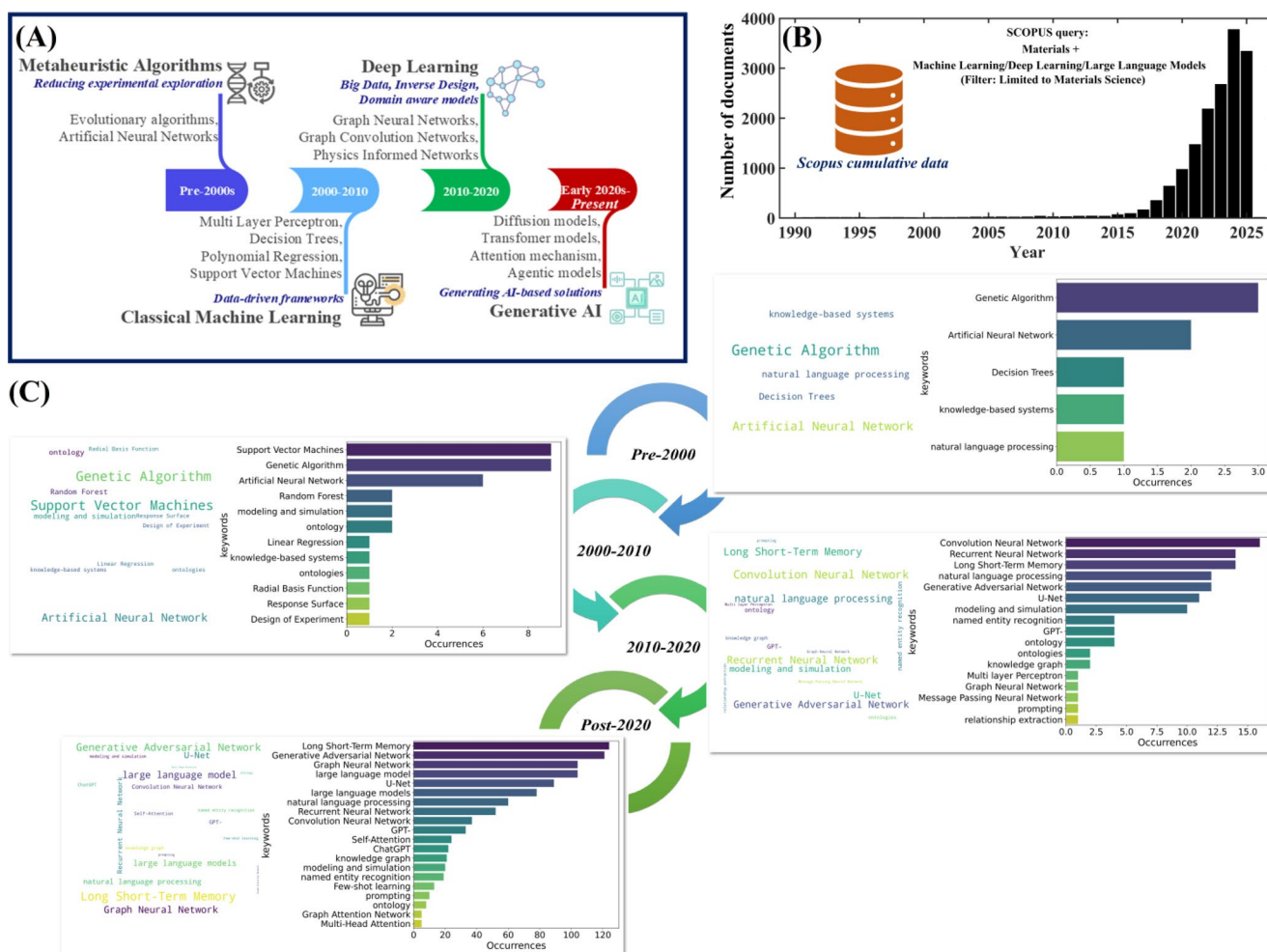


Fig. 12 Evolution and adoption of machine learning in advancing materials science. **(A)** Chronological trends in the integration of machine learning and artificial intelligence in materials science. **(B)** Publication statistics concerning the application of AI in the materials

domain. **(C)** Detailed distribution of the publication statistics on different subfields of machine learning and artificial intelligence concerning materials design and discovery

The following AI/ML keywords are used here, along with material-relevant keyword search:

Machine Learning: “Gaussian Processes”, “Linear Regression”, “Logistic Regression”, “Support Vector Machines”, “Decision Trees”, “Ensemble Trees”, “Random Forest”, “XGBoost”, “LightGBM”, “AdaBoost”, “CatBoost”, “Naive Bayes”, “K-Nearest Neighbors”, “Gaussian Process Regression”, “Active Learning”, “Bayesian Optimization”)

Deep Learning: “Artificial Neural Network”, “Multi-Layer Perceptron”, “Convolution Neural Network”, “Generative Adversarial Network”, “Long Short-Term Memory”, “Recurrent Neural Network”, “Graph Neural Network”, “Graph Convolution Network”, “Graph Attention Network”, “Message Passing Neural Network”, “Multi-Head Attention”, “Self-Attention”, “U-Net”, “Variational Auto Encoder”, “Physics Informed Neural Network”, “Neural

Operator”, “Fourier Neural Operator”, “Object Oriented”, “Multi Task Learning”, “Meta Learning”

Large Language Models: “Large Language Models”, “Large Language Model”, “ChatGPT”, “T5 Transformer”, “T5”, “BERT”, “GPT-”, “Few-Shot Learning”, “Knowledge-Based Systems”, “Natural Language Processing”, “Prompting”, “Knowledge Graph”, “Ontology”, “Modeling and Simulation”, “Event Based Simulations”, “Named Entity Recognition”, “Retrieval Augmented Generation”, “Reinforcement Learning from Human Feedback”, “Relationship Extraction”, “Relationship Classification”, “Agentic AI”, “One Shot”, “Few Shot”, “AI Agents”, “Materials Ontologies”, “Materials Ontology”, “Ontology”, “Ontologies”.

The use of artificial intelligence (AI) in materials design and discovery is rapidly evolving from predictive assistance

to an integrated, autonomous engine for scientific exploration. Initially, AI models in materials science focused on supervised learning tasks, such as predicting mechanical, electronic, or thermal properties from composition and structure. These models relied heavily on hand-engineered features and limited experimental or computational datasets. However, recent years have seen a shift toward more sophisticated AI approaches that enable inverse design, automated reasoning, and multi-modal data fusion.

One major trend is the application of generative models, such as Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), and Transformer architectures, that can propose entirely new material candidates with user-specified target properties. These models enable inverse design, allowing researchers to start with a desired property and work backwards to suggest structures or compositions that might exhibit it. Another growing area is multi-modal learning, where AI models integrate diverse data types, such as combining numerical simulation data, microscopy images, and scientific literature, to generate a richer understanding of materials behaviour. Additionally, self-supervised learning is being explored to make use of large amounts of unlabeled data, while foundation models, large AI models trained on vast corpora of materials knowledge, are emerging to offer general-purpose reasoning capabilities across different materials systems.

AI is increasingly being paired with active learning and closed-loop experimentation, where models iteratively guide experiments or simulations to efficiently explore material spaces. In parallel, physics-informed machine learning is being developed to embed scientific constraints, such as conservation laws or symmetry principles, into AI models to enhance their reliability and generalizability.

Scope for Improvement: Even with encouraging developments, there are still a number of issues that need to be resolved if artificial intelligence is to fully advance materials science. First, two of the biggest obstacles are data quality and availability. Large, high-quality datasets are necessary for the majority of machine learning models, but they are hard to come by in many material domains. Furthermore, there is little uniformity in the way that data is shared, annotated, and curated. Negative or failed experiments, which provide important information about what doesn't work, are rarely reported or used, and a large portion of the data produced in labs is unstructured, proprietary, or incomplete.

The current models frequently have trouble extrapolating and generalizing. When applied to materials that are similar to those in the training set, many AI models perform well; however, they are unable to accurately predict novel compositions, processing conditions, or material systems. This restricts their application in real-world discovery situations, where extrapolation beyond existing data is crucial.

Although they present viable solutions, strategies like transfer learning and meta-learning are still not widely used in this situation.

The interpretability and reliability of AI models continue to be major issues. Numerous deep learning techniques operate as “black boxes,” providing high accuracy without revealing the underlying physical principles. This presents a challenge for scientific research, where comprehension of mechanisms is just as crucial as forecasting. Techniques like explainable AI, symbolic regression, or causal inference frameworks that increase the transparency of AI models are desperately needed.

AI and lab experiments are frequently at odds. Very few research settings have real-time synthesis, processing, and characterization capabilities along with fully integrated AI. New automation tools, standardized data pipelines, and cooperative platforms that allow engineers, computer scientists, and materials scientists to jointly create intelligent experimental workflows will be necessary to close this gap.

AI's potential to produce an apparently plausible virtual reality is one of the main worries regarding its application in materials discovery across all length scales, which may or may not be true. If it is not correct, we will not always be able to verify for complex problems, and there will be cumulative errors to the extent that many things may break down due to the violation of basic laws of physics. Thus, multi-stage verification and infusion of physical insights to the AI models, wherever possible, should be the norm. A close collaboration of subject experts and AI developers will unlock the true potential of AI in materials design and discovery.

Emerging Technical Solutions to Data and Model Challenges in AI: The inherent problems in the materials-informatics regime of artificial intelligence consist of data scarcity (curse of data unavailability), low-resource data, lack of model generalization, and the black-box nature of developed models. These challenges most often hinder the seamless design and discovery of materials. Table 5 presents some of the popular methods to address these challenges.

Future Research Trajectories: Several key directions, most of which have already been incepted over the last couple of years, need focused attention from the research community to shape the future of AI in materials science in a productive way.

1. **Autonomous Materials Laboratories:** One major trajectory is the development of fully automated, AI-driven laboratories, often called “materials acceleration platforms” (MAPs) - that autonomously generate hypotheses, conduct experiments, analyze results, and refine

Table 5 List of solutions to data and model challenges with the example studies available in published scientific literature

Challenge	Solutions	Utilization in the materials domain
Data scarcity	Data augmentation (GAN, VAE)	• Generation of microstructural images using VAE-GAN [177] • Used GAN to generate the training dataset [178] • Generation of compositional data using VAE-GAN for predicting viscoelastic properties of asphalt binder [179]
	Transfer learning	• Used pretrained networks MobileNet, ResNet50 and VGG16 for transfer learning [177] • Use of weights of pretrained deep learning models for predicting the materials' performance for a small dataset [38] • Used the pretrained model XenonPy.MDL for making the predictions for a small-scale dataset [180]
	Active learning	• Use of a Bayesian active learning mechanism for developing a better ML model using a small dataset [51] • Use of random sampling based small dataset for developing efficient ML models [108]
Low-resource data	Data imputation	• Use of nearest neighbors for imputation of missing data in the materials domain [181, 182]
Models	Hybrid/Ensemble Methods	• Ensemble method to predict the drilling performance of composites [183] • Metamodelling approach for predicting the first-ply failure of composite [81]
Interpretable/ Explainable models	Explainable AI (XAI)	• SHAP-based post-hoc explanation of GPR model for predicting the mechanical performance of HEAs [109] • GradCAM-based explanation of identification of dislocation clusters in multicrystalline silicon [184]. • Local interpretation method for highlighting the text based on token importance for facilitating interpretation of the materials domain-aware language model [185].

models in real time. These systems could drastically reduce the time and cost required for materials discovery.

An important future research trajectory concerning machine learning potential [186] lies in the deeper integration of machine learning with molecular dynamics to enable adaptive, physics-aware simulation frameworks. ML models can learn reduced representations of configurational space, identify rare events, and guide on-the-fly sampling with quantified uncertainty, while selectively invoking DFT calculations for high-fidelity validation and data. The long-term direction is toward closed-loop ML–MD–DFT workflows, where interatomic potentials, sampling strategies, and accuracy control are updated autonomously, enabling scalable and reliable exploration of complex materials behaviour across extended time and length scales.

2. **Foundation Models for Materials Science:** Inspired by large language models like ChatGPT and image models like DALL·E, foundation models trained on massive, diverse materials datasets (text, images, simulations) may offer general-purpose reasoning, few-shot learning, and cross-domain insights. These models could support a wide range of applications, from literature mining to synthesis planning and property prediction.
3. **Sustainable and Circular Materials Design:** With growing attention to sustainability, AI will be increasingly applied to discover materials that are not only high-performing but also recyclable, low-impact, and aligned with circular economy principles. This includes models that can optimize materials for environmental as well as technical performance, incorporating life cycle analysis into the design loop.

4. **Multi-Scale Inverse Design:** Future research will move beyond atomic or molecular design toward multi-scale design, integrating composition, microstructure, processing history, and system-level performance. The evolving concepts of metamaterial designs across micro and nano scales, including multi-physical design concepts, will revolutionise the field of multi-scale inverse material designs [9, 187–194]. This shift will require new AI architectures capable of linking scales, from electrons to devices, structures and systems, for undertaking forward and inverse analyses, design and decision making.
5. **Ethics, Governance, and Open Science:** As AI becomes more central to materials innovation, issues of data governance, bias, reproducibility, and equity must be addressed. Community-driven efforts to develop open-source tools, FAIR datasets, and transparent benchmarks will be critical to ensure that AI advances benefit the broader scientific community.

In summary, AI is transforming materials discovery from a trial-and-error process to a more autonomous, hypothesis-driven discipline. While challenges remain, the integration of domain knowledge, advanced ML architectures, and real-time experimentation will define the next generation of materials innovation. Strategic focus on infrastructure, interdisciplinary collaboration, and open research culture will be key to unlocking the full potential of AI in materials science.

8 Conclusions

This article presents a critical review of the recent progress concerning the exploitation of artificial intelligence and machine learning for accelerated materials design and discovery. The emerging framework of materials-informatics is explained in the context of three different classes of materials - two-dimensional materials, composite materials and high entropy alloys. The chronological adoption and integration of artificial intelligence in the domain of materials science have been presented with critical insights, starting from supervised and unsupervised machine learning, deep learning, image-based learning algorithms and natural language processing, to generative and agentic AI. Evolving trends, scope of improvement, and future research trajectories are discussed based on the latest state of the literature concerning the concept of “materials by design”. In summary, this article offers a comprehensive overview of the materials-informatics pipeline, providing valuable perspectives on the end-to-end solution for materials design, uncovering the key concepts necessary for further developments in the field.

Acknowledgements TM would like to acknowledge the financial support received through an initiation grant from the University of Southampton during the period of this research.

Author Contributions All authors have contributed equally in preparing the manuscript.

Funding All necessary funding, including support for open access publication, is provided by University of Southampton.

Data Availability No datasets were generated or analysed during the current study.

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Rogers JA, DeSimone JM (2016) Novel materials. *Proc Natl Acad Sci* 113(42):11667–11669
- Hakim LK, Yazdani M, Alam M, Abbasi K, Tebyaniyan H, Tahmasebi E, Khayatan D, Seifalian A, Ranjbar R, Yazdani A (2021) Biocompatible and biomaterials application in drug delivery system in oral cavity. *Evid Based Complementary Altern Med* 2021(1):9011226
- Zou Y, Huang B, Cao L, Deng Y, Su J (2021) Tailored mesoporous inorganic biomaterials: assembly, functionalization, and drug delivery engineering. *Adv Mater* 33(2):2005215
- Yadav D, Garg RK, Ahlawat A, Chhabra D (2020) 3D printable biomaterials for orthopedic implants: solution for sustainable and circular economy. *Resour Policy* 68:101767
- Filip N, Radu I, Veliceasa B, Filip C, Perteu M, Clim A, Pinzariu AC, Drochioi IC, Hilitanu RL, Serban IL (2022) Biomaterials in orthopedic devices: current issues and future perspectives. *Coatings* 12(10):1544
- Wietelmann U, Klett J (2018) 200 years of lithium and 100 years of organolithium chemistry. *Z für Anorg und Allg Chem* 644(4):194–204
- Katz EA (2020) Perovskite: name puzzle and German-Russian odyssey of discovery. *Hel (Rome) Acta* 103(6):e2000061
- Wellmann PJ (2021) The search for new materials and the role of novel processing routes. *Discover Mater* 1(1):14
- Isanaka BR, Mukhopadhyay T, Varma RK, Kushvaha V (2022) On exploiting machine learning for failure pattern driven strength enhancement of honeycomb lattices. *Acta Materialia* 239:118226
- Bravyi S, Dial O, Gambetta JM, Gil D, Nazario Z (2022) The future of quantum computing with superconducting qubits. *J Appl Phys* 132(16)
- Choudhary K, DeCost B, Chen C, Jain A, Tavazza F, Cohn R, Park CW, Choudhary A, Agrawal A, Billinge SJ, Holm E (2022) Recent advances and applications of deep learning methods in materials science. *npj Comput Mater* 8(1):59
- Zaeem MA, Thomas S, Kavousi S, Zhang N, Mukhopadhyay T, Mahata A (2024) Multiscale computational modeling techniques in study and design of 2D materials: recent advances, challenges, and opportunities. *2D Mater* 11(4):042004
- Mahata A, Mukhopadhyay T, Chakraborty S, Asle Zaeem M (2024) Atomistic simulation assisted error-inclusive Bayesian machine learning for probabilistically unraveling the mechanical properties of solidified metals. *npj Comput Mater* 10(1):22
- Saumya K, Naskar S, Mukhopadhyay T (2023) ‘Magic’ of twisted multi-layered graphene and 2D nano-heterostructures. *Nano Futures* 7(3):032005
- de Pablo JJ, Jackson NE, Webb MA, Chen LQ, Moore JE, Morgan D, Jacobs R, Pollock T, Schlom DG, Toberer ES, Analytis J (2019) New frontiers for the materials genome initiative. *npj Comput Mater* 5(1):41
- Raccuglia P, Elbert KC, Adler PD, Falk C, Wenny MB, Mollo A, Zeller M, Friedler SA, Schrier J, Norquist AJ (2016) Machine-learning-assisted materials discovery using failed experiments. *Nature* 533(7601):73–76
- Jain A, Ong SP, Hautier G, Chen W, Richards WD, Dacek S, Cholia S, Gunter D, Skinner D, Ceder G, Persson KA (2013) Commentary: the materials Project: a materials genome approach to accelerating materials innovation. *APL Mater* 1(1)
- Ong SP, Cholia S, Jain A, Brafman M, Gunter D, Ceder G, Persson KA (2015) The materials application programming Interface (API): a simple, flexible and efficient API for materials data based on REpresentational state transfer (REST) principles. *Comput Mater Sci* 97:209–215
- Dung NQ, Khoi LV, Khoa NL, Hai V, Thanh DX, Anh P, Oanh NTH, Tuyen ND, Duc NM, Thang LH, Long PH (2024 October) Unveiling phase changes in Iron Ore reduction: X-Ray diffraction analysis with a user-friendly Colab Interface integrated with the materials Project API. In: The International Conference on

- Advanced Materials and Technology (ICAMT 2024), Hanoi University of Science and Technology.
20. Liu H, Cheng J, Dong H, Feng J, Pang B, Tian Z, Ma S, Xia F, Zhang C, Dong L (2020) Screening stable and metastable ABO₃ perovskites using machine learning and the materials project. *Comput Mater Sci* 177:109614
 21. Haastrup S, Strange M, Pandey M, Deilmann T, Schmidt PS, Hinsche NF, Gjerding MN, Torelli D, Larsen PM, Riis-Jensen AC, Gath J (2018) The computational 2D materials database: high-throughput modeling and discovery of atomically thin crystals. *2D Mater* 5(4):042002
 22. Gražulis S, Merkys A, Vaitkus A (2020) Crystallography open database (COD). *Handbook of materials modeling: Methods: Theory and modeling*, pp 1863–1881
 23. Saal JE, Kirklin S, Aykol M, Meredig B, Wolverton C (2013) Materials design and discovery with high-throughput density functional theory: the open quantum materials database (OQMD). *Jom* 65:1501–1509
 24. Venkatraman V, Carvalho PA (2022) On the value of popular crystallographic databases for machine learning prediction of space groups. *Acta Materialia* 240:118353
 25. Veit-Acosta M, de Azevedo Junior WF (2021) The impact of crystallographic data for the development of machine learning models to predict protein-ligand binding affinity. *Curr Med Chem* 28(34):7006–7022
 26. Souza A, Oliveira LB, Hollatz S, Feldman M, Olukotun K, Holton JM, Cohen AE, Nardi L (2019) DeepFreak: learning crystallography diffraction patterns with automated machine learning. *arXiv preprint arXiv: 1904.11834*
 27. Ward L, Dunn A, Faghaninia A, Zimmermann NE, Bajaj S, Wang Q, Montoya J, Chen J, Bystrom K, Dylla M, Chard K (2018) Matminer: an open source toolkit for materials data mining. *Comput Mater Sci* 152:60–69
 28. Islam MM (2024) Low dimensional fragment-based descriptors for property predictions in inorganic materials with machine learning. *arXiv preprint arXiv: 2407.21146*
 29. Yazdani MH, Branicio PS, Nomura KI (2024) Benchmarking machine learning models on a dielectric constant database for bandgap prediction
 30. Mazumder DD, Islam T, Ushna AN (2024) Band gap prediction of inorganic materials using artificial intelligence through feature transformation. Available at SSRN 4857921
 31. Jiayuan D, Xiaoyu Y (2020) Integration and optimization of material data mining and machine learning tools. *Front Data Computing* 2(4):105–120
 32. Ong SP, Richards WD, Jain A, Hautier G, Kocher M, Cholia S, Gunter D, Chevrier VL, Persson KA, Ceder G (2013) Python materials genomics (pymatgen): a robust, open-source python library for materials analysis. *Comput Mater Sci* 68:314–319
 33. Peivaste I, Ramezani S, Alahyarizadeh G, Ghaderi R, Makradi A, Belouettar S (2024) Rapid and accurate predictions of perfect and defective material properties in atomistic simulation using the power of 3D CNN-based trained artificial neural networks. *Sci Rep* 14(1):36
 34. Zhao H, Ezech CI, Ren W, Li W, Pang CH, Zheng C, Gao X, Wu T (2019) Integration of machine learning approaches for accelerated discovery of transition-metal dichalcogenides as Hg₀ sensing materials. *Appl Energy* 254:113651
 35. Ong SP (2019) Accelerating materials science with high-throughput computations and machine learning. *Comput Mater Sci* 161:143–150
 36. Chen C, Nguyen DT, Lee SJ, Baker NA, Karakoti AS, Lauw L, Owen C, Mueller KT, Bilodeau BA, Murugesan V, Troyer M (2024) Accelerating computational materials discovery with machine learning and cloud high-performance computing: from large-scale screening to experimental validation. *J Am Chem Soc* 146(29):20009–20018
 37. Lambrix P, Armiento R, Li H, Hartig O, Abd Nikoie Pour M, Li Y (2024) The materials design ontology. *Semant Web* 15(2):481–515
 38. Gupta V, Choudhary K, Tavazza F, Campbell C, Liao WK, Choudhary A, Agrawal A (2021) Cross-property deep transfer learning framework for enhanced predictive analytics on small materials data. *Nat Commun* 12(1):6595
 39. Curtarolo S, Setyawan W, Hart GL, Jahnatek M, Chepulskii RV, Taylor RH, Wang S, Xue J, Yang K, Levy O, Mehl MJ (2012) Aflow: an automatic framework for high-throughput materials discovery. *Comput Mater Sci* 58:218–226
 40. Roberts J, Rijal B, Divilov S, Maria JP, Fahrenholtz WG, Wolfe DE, Brenner DW, Curtarolo S, Zurek E (2024) Machine learned interatomic potentials for ternary carbides trained on the aflow database. *npj Comput Mater* 10(1):142
 41. Sbailò L, Fekete Á, Ghiringhelli LM, Scheffler M (2022) The nomad artificial-intelligence toolkit: turning materials-science data into knowledge and understanding. *npj Comput Mater* 8(1):250
 42. Foppa L, Purcell TA, Levchenko SV, Scheffler M, Ghiringhelli LM (2022) Hierarchical symbolic regression for identifying key physical parameters correlated with bulk properties of perovskites. *Phys Rev Lett* 129(5):055301
 43. M. Hutchinson, S. Paradiso, and L. Ward, *Citrine informatics* lolo, 2016, available at <https://github.com/CitrineInformatics/lolo>, accessed: August 27, 2019
 44. Fong AY, Pellouchoud L, Davidson M, Walroth RC, Church C, Tcareva E, Wu L, Peterson K, Meredig B, Tassone CJ (2021) Utilization of machine learning to accelerate colloidal synthesis and discovery. *J Chem Phys* 154(22)
 45. Choudhary K, Garrity KF, Reid AC, DeCost B, Biacchi AJ, Walker ARH, Trautt Z, Hattrick-Simpers J, Kusne AG, Centrone A et al (2020) The joint automated repository for various integrated simulations (jarvis) for data-driven materials design, *npj computational materials* 6(1):173
 46. Kaundinya PR, Choudhary K, Kalidindi SR (2022) Prediction of the electron density of states for crystalline compounds with atomistic line Graph neural networks (ALIGNNN). *Jom* 74(4):1395–1405
 47. Zhao X, Liu Z, Zhao L, Gao XW, Ren T, Luo WB (2025) Machine learning helps data mining to build descriptor databases for lithium-ion batteries. *MetalMat* e70008
 48. Weston L, Tshitoyan V, Dagdelen J, Kononova O, Trewartha A, Persson KA, Ceder G, Jain A (2019) Named entity recognition and normalization applied to large-scale information extraction from the materials science literature. *J Chem Inf Modeling* 59(9):3692–3702
 49. Huang Z, He L, Yang Y, Li A, Zhang Z, Wu S, Wang Y, He Y, Liu X (2024) Application of machine reading comprehension techniques for named entity recognition in materials science. *J Cheminf* 16(1):76
 50. Zhang Y, Xu W, Liu G, Zhang Z, Zhu J, Li M (2021) Bandgap prediction of two-dimensional materials using machine learning. *PLoS One* 16(8):e0255637
 51. Mohanty T, Chandran KS, Sparks TD (2023) Machine learning guided optimal composition selection of niobium alloys for high temperature applications. *APL Mach Learn* 1(3)
 52. Bazgir A, Zhang Y (2025) Multicrossmodal automated agent for Integrating diverse materials science data. *arXiv preprint arXiv: 2505.15132*
 53. Barman S, Kumar Gupta K, Dey S (2024) Exploring nano-scale scratching induced tribological behavior of graphene engineered AlCoCrFeNi high-entropy alloy. *J Appl Mech* 91(10)

54. Barman S, Gupta KK, Dey S (2024) Probing atomistic deformation behavior of graphene-coated Al₁₀. 3CoCrFeNi high-entropy alloy under nanoindentation. *J Micromech Mol Phys* 1–14
55. Barman S, Gupta KK, Dey S (2024) Enhancing mechanical performance of Al₁₀. 3CoCrFeNi HEA films through graphene coating: insights from nanoindentation and dislocation mechanism analysis. *Modell Simul Mater Sci Eng* 32(3):035012
56. Roy A, Gupta KK, Dey S, Mukhopadhyay T (2024) Random topological defects in double-walled carbon nanotubes: on characterization and programmable defect-engineering of spatio-mechanical properties. *Adv Nano Res* 16(1):91
57. Gupta KK, Mukhopadhyay T, Dey S (2023) Probing the molecular-level energy absorption mechanism and strategic sequencing of graphene/Al composite laminates under high-velocity ballistic impact of nano-projectiles. *Appl Surf Sci* 629:156502
58. Gupta KK, Mukhopadhyay T, Roy A, Dey S (2020) Probing the compound effect of spatially varying intrinsic defects and doping on mechanical properties of hybrid graphene monolayers. *J Mater Sciamp; Technol* 50:44–58
59. Liu Q, Chen W, Yakubov V, Kruzic JJ, Wang CH, Li X (2024) Interpretable machine learning approach for exploring process-structure-property relationships in metal additive manufacturing. *Addit. Manuf* 85 [online]
60. Thoppil GS, Nie JF, Alankar A (2023) Bayesian approach for inferable machine learning models of process-structure-property linkages in complex concentrated alloys. *J Alloys Compd* 967:171595
61. Tran H, Kim C, Gurnani R, Hvidsten O, DeSimpliciis J, Ramprasad R, Gadelrab K, Tuffile C, Molinari N, Kitchaev D, Kornbluth M (2024) Polymer composites informatics for flammability, thermal. *Mech Electr Property Predictions*. arXiv preprint arXiv: 2412.08407
62. Alberi K, Nardelli MB, Zakutayev A, Mitas L, Curtarolo S, Jain A, Fornari M, Marzari N, Takeuchi I, Green ML, Kanatzidis M (2018) The 2019 materials by design roadmap. *J Phys D: Appl Phys* 52(1):013001. CC BY 3.0. <https://doi.org/10.1088/1361-6463/aad7aa>
63. Frey NC, Akinwande D, Jariwala D, Shenoy VB (2020) Machine learning-enabled design of point defects in 2D materials for quantum and neuromorphic information processing. *ACS Nano* 14(10):13406–13417
64. Frey NC, Wang J, Vega Bellido GI, Anasori B, Gogotsi Y, Shenoy VB (2019) Prediction of synthesis of 2D metal carbides and nitrides (MXenes) and their precursors with positive and unlabeled machine learning. *ACS Nano* 13(3):3031–3041
65. Kim HJ, Chong M, Rhee TG, Khim YG, Jung MH, Kim YM, Jeong HY, Choi BK, Chang YJ (2023) Machine-learning-assisted analysis of transition metal dichalcogenide thin-film growth. *Nano Convergence* 10(1):10
66. Manti S, Svendsen MK, Knøsgaard NR, Lyngby PM, Thygesen KS (2023) Exploring and machine learning structural instabilities in 2D materials. *npj Comput Mater* 9(1):33
67. Choudhary K, Garrity KF, Hartman ST, Pilania G, Tavazza F (2023) Efficient computational design of two-dimensional van der waals heterostructures: band alignment, lattice mismatch, and machine learning. *Phys Rev Mater* 7(1):014009
68. Pang Y, Yang Z, Yang Y, Ren TL (2020) Wearable electronics based on 2D materials for human physiological information detection. *Small* 16(15):1901124
69. Yang T, Jiang X, Huang Y, Tian Q, Zhang L, Dai Z, Zhu H (2022) mechanical sensors based on two-dimensional materials: sensing mechanisms, structural designs and wearable applications. *Isience* 25(1)
70. Gupta KK, Mukhopadhyay T, Roy A, Roy L, Dey S (2021) Sparse machine learning assisted deep computational insights on the mechanical properties of graphene with intrinsic defects and doping. *J Phys Chem Solids* 155:110111
71. Gupta KK, Roy L, Dey S (2022) Ballistic performance of Bi-layer graphene: artificial neural network based molecular dynamics simulations. In: *Machine learning applied to composite materials*. Springer Nature Singapore, Singapore, pp 135–148
72. Gupta KK, Roy A, Mukhopadhyay T, Roy L, Dey S (2022) Probing the stochastic fracture behavior of twisted bilayer graphene: efficient ann based molecular dynamics simulations for complete probabilistic characterization. *Mater Today Commun* 32:103932
73. Gupta KK, Roy L, Dey S (2022) Hybrid machine-learning-assisted stochastic nano-indentation behaviour of twisted bilayer graphene. *J Phys Chem Solids* 167:110711
74. Roy A, Gupta KK, Dey S (2022) Probabilistic investigation of temperature-dependent vibrational behavior of hetero-nanotubes. *Appl Nanosci* 12(7):2077–2089
75. Gupta KK, Mukhopadhyay T, Roy L, Dey S (2022) High-velocity ballistics of twisted bilayer graphene under stochastic disorder. *Adv Nano Res* 12(5):529–547
76. Gupta KK, Roy L, Dey S (2022) Machine learning-based molecular dynamics simulations of monolayered graphene. In *Recent Advances in Computational and Experimental Mechanics, Vol II: Select Proceedings of ICRACEM 2020*, Springer Nature Singapore, Singapore, 251–263
77. Chang J, Zhu S (2024) Deep learning on atomistic physical fields of graphene for strain and defect Engineering. *Adv Intell Syst* 6(4):2300601
78. Shishir MIR, Elapolu MSR, Tabarraei A (2023) A deep learning model for predicting mechanical properties of polycrystalline graphene. *Comput Mater Sci* 218:111924
79. Dewapriya MAN, Rajapakse RKND, Dias WPS (2020) Characterizing fracture stress of defective graphene samples using shallow and deep artificial neural networks. *Carbon* 163:425–440
80. Dey S, Mukhopadhyay T, Adhikari S (2017) Metamodel based high-fidelity stochastic analysis of composite laminates: a concise review with critical comparative assessment. *Composite Struct* 171:227–250
81. Kushari S, Gupta KK, Vaishali, Dey S (2022) Metamodeling-assisted probabilistic first ply failure analysis of laminated composite plates-RS-HDMR-and GPR-based approach. *J Braz Soc Of Mech Sci And Eng* 44(8):374
82. Raturi HP, Kushari S, Karsh PK, Dey S (2024) Evaluating stochastic Fundamental natural frequencies of porous functionally graded material plate with even porosity effect: a multi-machine learning approach. *J Vib Eng Technol* 12(2):1931–1942
83. Vaishali K, Kushari PK, Kumar S, RR, Dey S (2023) Stochastic free vibration and impact responses of functionally graded plates: a support vector machine learning model approach. *J Vib Eng Technol* 11(7):2927–2943
84. Mukhopadhyay T, Karsh PK, Basu B, Dey S (2020) Machine learning based stochastic dynamic analysis of functionally graded shells. *Composite Struct* 237:111870
85. Divakaraju PV, Mishra N, Pandurangan V, Nithyadharan M (2023) Machine learning based inverse framework for predicting the transverse and shear modulus of carbon fiber. *Comput Mater Sci* 230:112518
86. Yossef M, Noureldin M, Alqabbany A (2024 Aug 1) Explainable artificial intelligence framework for FRP composites design. *Composite Struct* 341:118190
87. Yao L, Wang J, Chuai M, Lomov SV, Carvelli V (2024) Physics-informed machine learning for loading history dependent fatigue delamination of composite laminates. *Compos Part A: Appl Sci Manuf* 187:108474
88. Bharadwaja BVSS, Nabian MA, Sharma B, Choudhry S, Alankar A (2022) Physics-informed machine learning and uncertainty

- quantification for mechanics of heterogeneous materials. Integrating Mater Manuf Innov 11(4):607–627
89. Zhang B, Shi D, Liu C, Hao W, Yang X (2023) Physics-informed machine learning with high-throughput design module for evaluating rupture life and guiding design of oxide/oxide ceramic matrix composites. *Ceram Int* 49(22):34945–34957
 90. Khan A, Ko DK, Lim SC, Kim HS (2019) Structural vibration-based classification and prediction of delamination in smart composite laminates using deep learning neural network. *Compos Part B: Eng* 161:586–594
 91. Yan H, Xie W, Gao B, Yang F, Meng S (2023) A multi-task learning model for fast prediction of mechanical behavior of UD-CFRP composites under transverse tension. *Composite Struct* 324:117555
 92. Zhou K, Sun H, Enos R, Zhang D, Tang J (2021) Harnessing deep learning for physics-informed prediction of composite strength with microstructural uncertainties. *Comput Mater Sci* 197:110663
 93. Bhaduri A, Gupta A, Graham-Brady L (2022) Stress field prediction in fiber-reinforced composite materials using a deep learning approach. *Compos Part B: Eng* 238:109879
 94. Gupta S, Mukhopadhyay T, Kushvaha V (2023) Microstructural image based convolutional neural networks for efficient prediction of full-field stress maps in short fiber polymer composites. *Def Technol* 24:58–82
 95. Gupta S, Mukhopadhyay T, Varade D, Kushvaha V (2023 December) Investigating deep learning-based stress prediction in particulate polymer composites using multiple quality measures. In *International Conference on Fundamental and Industrial Research in Materials*, Springer Nature Singapore, Singapore, 101–107
 96. Krishna SA, Noble N, Radhika N, Saleh B (2024) A comprehensive review on advances in high entropy alloys: fabrication and surface modification methods, properties, applications, and future prospects. *J Manuf Processes* 109:583–606
 97. Sonar T, Ivanov M, Trofimov E, Tingaev A, Suleymanova I (2024) An overview of microstructure, mechanical properties and processing of high entropy alloys and its future perspectives in aeroengine applications. *Mater Sci For Energy Technol* 7:35–60
 98. Li J, Ye Z, Fu J, Qi W, Tian Y, Liu L, Wang X (2020) Microstructure evolution, texture and laser surface HEACs of Al-mg-Si alloy for light automobile parts. *Mater Charact* 160:110093
 99. Moschetti M, Burr PA, Obbard E, Kruzic JJ, Hosemann P, Gludovatz B (2022) Design considerations for high entropy alloys in advanced nuclear applications. *J Nucl Mater* 567:153814
 100. Shargh AK, Stiles CD, El-Awady JA (2024) Deep learning accelerated phase prediction of refractory multi-principal element alloys. *Acta Materialia* 120558
 101. Shargh AK, Stiles CD, El-Awady JA (2024) Temperature-dependent design of BCC high entropy alloys: integrating deep learning, CALPHAD, and experimental validation. *arXiv preprint arXiv: 2410.14614*
 102. Rafiei MH, Gu Y, El-Awady JA (2020) Machine learning of dislocation-induced stress fields and interaction forces. *JOM* 72(12):4380–4392
 103. Revi V, Kasodariya S, Talapatra A, Pilania G, Alankar A (2021) Machine learning elastic constants of multi-component alloys. *Comput Mater Sci* 198:110671
 104. Beausoleil GL, Aguiar JA, Kwon S, Parry ME, Beatty D, Curnutt BJ, Sparks T, Eyerman E (2020) Decoding early candidacy of high entropy alloys for nuclear application using the advanced Test reactor through predictive methods and combinatorial testing No. INL/CON-20-57066-Rev. 000. Idaho National Lab (INL), Idaho Falls, ID (United States).
 105. Zhang L, Qian K, Huang J, Liu M, Shibuta Y (2021) Molecular dynamics simulation and machine learning of mechanical response in non-equiatom FeCrNiCoMn high-entropy alloy. *J Mater Res Technol* 13:2043–2054
 106. Jha D, Ward L, Paul A, Liao WK, Choudhary A, Wolverson C, Agrawal A (2018) Elemnet: deep learning the chemistry of materials from only elemental composition. *Sci Rep* 8(1):17593
 107. Wen C, Zhang Y, Wang C, Xue D, Bai Y, Antonov S, Dai L, Lookman T, Su Y (2019) Machine learning assisted design of high entropy alloys with desired property. *Acta Materialia* 170:109–117
 108. Gupta KK, Barman S, Dey S, Naskar S, Mukhopadhyay T (2024) On exploiting nonparametric kernel-based probabilistic machine learning over the large compositional space of high entropy alloys for optimal nanoscale ballistics. *Sci Rep* 14(1):16795
 109. Gupta KK, Barman S, Dey S, Mukhopadhyay T (2024) Explainable machine learning assisted molecular-level insights for enhanced specific stiffness exploiting the large compositional space of AlCoCrFeNi high entropy alloys. *Mach Learn Sci Technol*
 110. Imam F, Musilek P, Reformat MZ (2024) Parametric and non-parametric machine learning techniques for increasing power system reliability: a review. *Information* 15(1):37
 111. Su X, Yan X, Tsai CL (2012) Linear regression. *Wiley Interdiscip Rev: Comput Stat* 4(3):275–294
 112. Ostertagová E (2012) Modelling using polynomial regression. *Procedia Eng* 48:500–506
 113. LaValley MP (2008) Logistic regression. *Circulation* 117(18):2395–2399
 114. Krishnan NA (2024) Machine learning for materials discovery: numerical recipes and practical applications. Springer Nature
 115. Samet H (2007) K-nearest neighbor finding using MaxNearest-Dist. *IEEE Trans Pattern Anal Mach Intel* 30(2):243–252
 116. Suthaharan S, Suthaharan S (2016) Support vector machine. *machine learning models and algorithms for big data classification: thinking with examples for effective learning* 207–235
 117. De Ville B (2013) Decision trees. *Wiley Interdiscip Rev: Comput Stat* 5(6):448–455
 118. Schulz E, Speekenbrink M, Krause A (2018) A tutorial on Gaussian process regression: modelling, exploring, and exploiting functions. *J Math Psychol* 85:1–16
 119. LeCun Y, Bengio Y, Hinton G (2015) Deep learning. *Nature* 521(7553):436–444
 120. Li Z, Liu F, Yang W, Peng S, Zhou J (2021) A survey of convolutional neural networks: analysis, applications, and prospects. *IEEE Trans Neural Networks Learn Syst* 33(12):6999–7019
 121. LeNail (2019) NN-SVG: publication-ready neural network architecture schematics. *J Open Source Softw* 4(33):747. <https://doi.org/10.21105/joss.00747>
 122. Gavrikov P (2020–2024 December) *visualkeras*. github repository. Available at: <https://github.com/paulgavrikov/visualkeras> (Accessed:20)
 123. Alom MZ, Taha TM, Yakopcic C, Westberg S, Sidike P, Nasrin MS, Van Esesn BC, Awwal AAS, Asari VK (2018) The history began from alexnet: a comprehensive survey on deep learning approaches. *arXiv preprint arXiv: 1803.01164*
 124. Tammina S (2019) Transfer learning using vgg-16 with deep convolutional neural network for classifying images. *Int J Sci And Res Publications (IJSRP)* 9(10):143–150
 125. Targ S, Almeida D, Lyman K (2016) Resnet in resnet: generalizing residual architectures. *arXiv preprint arXiv: 1603.08029*
 126. Chen LC, Zhu Y, Papandreou G, Schroff F, Adam H (2018) Encoder-decoder with atrous separable convolution for semantic image segmentation. In *Proceedings of the European conference on computer vision (ECCV)*, pp 801–818
 127. Kar MK, Nath MK (2024) Efficient segmentation of vessels and disc simultaneously using multi-channel generative adversarial network. *SN Comput Sci* 5(3):288

128. Variyar VS, Sowmya V, Sivanpillai R, Brown GK (2024) A multi-branch dual attention segmentation network for epiphyte drone images. *Image Vision Comput* 105099
129. Vv SV, Sowmya V, Sivanpillai R, Brown GK (2024) Significance of AI-assisted techniques for epiphyte plant monitoring and identification from drone images. *J Environ Manag* 367:121996
130. De Mulder W, Bethard S, Moens MF (2015) A survey on the application of recurrent neural networks to statistical language modeling. *Comput Speech Lang* 30(1):61–98
131. Dey R, Salem FM (2017 August) Gate-variants of gated recurrent unit (gru) neural networks. In 2017 IEEE 60th international midwest symposium on circuits and systems (MWSCAS), IEEE (pp. 1597–1600
132. Ghosh S, Vinyals O, Strobe B, Roy S, Dean T, Heck L (2016) Contextual lstm (clstm) models for large scale nlp tasks. *arXiv preprint arXiv: 1602.06291*
133. Vaswani A (2017) Attention is all you need. *Adv Neural Inf Process Syst*
134. Ghodsi A, Ca U Attention mechanism. Tutorial and survey, Transformers, BERT, and GPT
135. Shetty P, Rajan AC, Kuenneth C, Gupta S, Panchumarti LP, Holm L, Zhang C, Ramprasad R (2023) A general-purpose material property data extraction pipeline from large polymer corpora using natural language processing. *npj Comput Mater* 9(1):52. CC BY 4.0. <https://doi.org/10.1038/s41524-023-00980-3>
136. Gupta T, Zaki M, Krishnan NA and Mausam (2022) MatSciBERT: a materials domain language model for text mining and information extraction. *npj Comput Mater* 8(1):102.
137. Wei L, Li Q, Song Y, Stefanov S, Dong R, Fu N, Siriwardane EM, Chen F, Hu J (2024) Crystal composition transformer: self-learning neural language model for generative and tinkering design of materials. *Adv Sci* 11(36):2304305
138. Durant JL, Leland BA, Henry DR, Nourse JG (2002) Reoptimization of MDL keys for use in drug discovery. *J Chem Inf Comput Sci* 42:1273–1280
139. Butler KT, Davies DW, Cartwright H, Isayev O, Walsh A (2018) Machine learning for molecular and materials science. *Nature* 559(7715):547–555
140. Ramprasad R, Batra R, Pailania G, Mannodi-Kanakkithodi A, Kim C (2017) Machine learning in materials informatics: recent applications and prospects. *npj Comput Mater* 3(1):54
141. Ward L, Liu R, Krishna A, Hegde VI, Agrawal A, Choudhary A, Wolverton C (2017) Including crystal structure attributes in machine learning models of formation energies via Voronoi tessellations. *Phys Rev B* 96(2):024104
142. Pailania G, Wang C, Jiang X, Rajasekaran S, Ramprasad R (2013) Accelerating materials property predictions using machine learning. *Sci Rep* 3(1):2810
143. Tshitoyan V, Dagdelen J, Weston L, Dunn A, Rong Z, Kononova O, Persson KA, Ceder G, Jain A (2019) Unsupervised word embeddings capture latent knowledge from materials science literature. *Nature* 571(7763):95–98
144. Huo H, Rupp M (2022) Unified representation of molecules and crystals for machine learning. *Mach Learn: Sci Technol* 3(4):045017
145. Ghiringhelli LM, Vybiral J, Levchenko SV, Draxl C, Scheffler M (2015) Big data of materials science: critical role of the descriptor. *Phys Rev Lett* 114(10):105503
146. Faber FA, Hutchison L, Huang B, Gilmer J, Schoenholz SS, Dahl GE, Vinyals O, Kearnes S, Riley PF, Von Lilienfeld OA (2017) Prediction errors of molecular machine learning models lower than hybrid DFT error. *J Chem Theory Computation* 13(11):5255–5264
147. Schütt KT, Arbabzadah F, Chmiela S, Müller KR, Tkatchenko A (2017) Quantum-chemical insights from deep tensor neural networks. *Nat Commun* 8(1):13890
148. DeCost BL, Francis T, Holm EA (2017) Exploring the microstructure manifold: image texture representations applied to ultrahigh carbon steel microstructures. *Acta Materialia* 133:30–40
149. Xie T, Grossman JC (2018) Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. *Phys Rev Lett* 120(14):145301
150. Noé F, Tkatchenko A, Müller KR, Clementi C (2020) Machine learning for molecular simulation. *Annu Rev Phys Chem* 71(1):361–390
151. Gu Y, Tinn R, Cheng H, Lucas M, Usuyama N, Liu X, Naumann T, Gao J, Poon H (2021) Domain-specific language model pre-training for biomedical natural language processing. *ACM Trans Comput Healthcare (health)* 3(1):1–23
152. Beltagy I, Lo K, Cohan A (2019) SciBERT: a pretrained language model for scientific text. *arXiv preprint arXiv: 1903.10676*
153. Huang H, Magar R, Xu C, Farimani AB (2023) Materials informatics transformer: a language model for interpretable materials properties prediction. *arXiv preprint arXiv: 2308.16259*
154. Trewartha A, Walker N, Huo H, Lee S, Cruse K, Dagdelen J, Dunn A, Persson KA, Ceder G, Jain A (2022) Quantifying the advantage of domain-specific pre-training on named entity recognition tasks in materials science. *Patterns* 3(4)
155. Choudhary K (2024) Atomgpt: atomistic generative pretrained transformer for forward and inverse materials design. *J Phys Chem Lett* 15(27):6909–6917
156. Prasad D, Pimpude M, Alankar A (2024) Towards development of automated knowledge maps and databases for materials Engineering using large language models. *arXiv preprint arXiv: 2402.11323*
157. Wang AYT, Kauwe SK, Murdock RJ, Sparks TD (2021) Compositionally restricted attention-based network for materials property predictions. *npj Comput Mater* 7(1):77
158. Chandrasekhar A, Chan J, Ogoke F, Ajenifujah O, Farimani AB (2024) AMGPT: a large language model for contextual querying in additive Manufacturing. *arXiv preprint arXiv: 2406.00031*
159. Sayeed HM, Smallwood W, Baird SG, Sparks TD (2024) NLP meets materials science: quantifying the presentation of materials data in literature. *Matter* 7(3):723–727
160. Sun Y, Li X, Liu C, Deng X, Zhang W, Wang J, Zhang Z, Wen T, Song T, Ju D (2024) Development of an intelligent design and simulation aid system for heat treatment processes based on LLM. *Mater & Des* 248:113506
161. Kapoor G, Chawla K, Ghosal T, Villez K, Coughlin D, Rucker T, Paquit V, Ozcan S, Kim S (2025 June) Intelligent Manufacturing support: specialized LLMs for composite material processing and equipment operation. In International Manufacturing Science and Engineering Conference, vol 89022. American Society of Mechanical Engineers, p. V002T20A009
162. Honda S, Shi S, Ueda HR (2019) Smiles transformer: pre-trained molecular fingerprint for low data drug discovery. *arXiv preprint arXiv: 1911.04738*
163. Irwin R, Dimitriadis S, He J, Bjerrum EJ (2022) Chemformer: a pre-trained transformer for computational chemistry. *Mach Learn: Sci Technol* 3(1):015022
164. Anselmi M, Slabaugh G, Crespo-Otero R, Di Tommaso D (2024) Molecular graph transformer: stepping beyond ALIGNN into long-range interactions. *Digit Discov* 3(5):1048–1057
165. Kim S, Noh J, Gu GH, Aspuru-Guzik A, Jung Y (2020) Generative adversarial networks for crystal structure prediction. *ACS Central Sci* 6(8):1412–1420
166. Fokina D, Muravleva E, Ovchinnikov G, Oseledets I (2020) Microstructure synthesis using style-based generative adversarial networks. *Phys Rev E* 101(4):043308
167. Schwaller P, Laino T, Gaudin T, Bolgar P, Hunter CA, Bekas C, Lee AA (2019) Molecular transformer: a model for

- uncertainty-calibrated chemical reaction prediction. *ACS Central Sci* 5(9):1572–1583
168. Sanchez-Lengeling B, Aspuru-Guzik A (2018) Inverse molecular design using machine learning: generative models for matter engineering. *Science* 361(6400):360–365
 169. Karniadakis GE, Kevrekidis IG, Lu L, Perdikaris P, Wang S, Yang L (2021) Physics-informed machine learning. *Nat Rev Phys* 3(6):422–440
 170. Rohr B, Stein HS, Guevarra D, Wang Y, Haber JA, Aykol M, Suram SK, Gregoire JM (2020) Benchmarking the acceleration of materials discovery by sequential learning. *Chem Sci* 11(10):2696–2706
 171. Gridach M, Nanavati J, Abidine KZE, Mendes L, Mack C (2025) Agentic ai for scientific discovery: a survey of progress, challenges, and future directions. *arXiv preprint arXiv: 2503.08979*
 172. Feng R, Liang Y, Yin T, Gao P, Wang W (2025) Agentic Assistant for materials scientists. *The Electrochem Soc Interface* 34(2):45
 173. Van MH, Verma P, Zhao C, Wu X (2025) A survey of AI for materials science: foundation models, LLM agents, datasets, and tools. *arXiv preprint arXiv: 2506.20743*
 174. Ghafarollahi A, Buehler MJ (2025) Automating alloy design and discovery with physics-aware multimodal multiagent AI. *Proc Natl Acad Sci* 122(4):e2414074122
 175. Hung L, Yager JA, Monteverde D, Baiocchi D, Kwon HK, Sun S, Suram S (2024) Autonomous laboratories for accelerated materials discovery: a community survey and practical insights. *Digit Discov* 3(7):1273–1279
 176. Green ML, Maruyama B, Schrier J (2022) Autonomous (AI-driven) materials science. *Appl Phys Rev* 9(3)
 177. Han Y, Liu Y, Chen Q (2023) Data augmentation in material images using the improved HP-VAE-GAN. *Comput Mater Sci* 226:112250
 178. Ohno H (2022) Training data augmentation using generative models with statistical guarantees for materials informatics. *Soft Comput* 26(3):1181–1196
 179. Wang Y, Shan L, Yan J, Pang Y (2025) A data augmentation method for establishing a relationship model between composition and viscoelastic properties of Asphalt Binder. *IEEE Trans Intell Transp Syst*
 180. Yamada H, Liu C, Wu S, Koyama Y, Ju S, Shiomi J, Morikawa J, Yoshida R (2019) Predicting materials properties with little data using shotgun transfer learning. *ACS Central Sci* 5(10):1717–1730
 181. Xie C, Li R, Li Y, Xie H, Liu Q (2024) Imputation of missing data in materials science through nearest neighbors and iterative predictions. *J Chem Theory Computation* 21(1):70–78
 182. Lyngdoh GA, Zaki M, Krishnan NA, Das S (2022) Prediction of concrete strengths enabled by missing data imputation and interpretable machine learning. *Cem Concr Compos* 128:104414
 183. Baraheni M, Soudmand BH, Amini S, Fotouhi M (2024) Stacked generalization ensemble learning strategy for multivariate prediction of delamination and maximum thrust force in composite drilling. *J Composite Mater* 58(30):3113–3138
 184. Hara K, Kojima T, Kutsukake K, Kudo H, Usami N (2023) 3D CNN and grad-CAM based visualization for predicting generation of dislocation clusters in multicrystalline silicon. *APL Mach Learn* 1(3)
 185. Korolev V, Protsenko P (2023) Accurate, interpretable predictions of materials properties within transformer language models. *Patterns* 4(10)
 186. Mortazavi B, Zhuang X, Rabczuk T, Shapeev AV (2023) Atomistic modeling of the mechanical properties: the rise of machine learning interatomic potentials. *Material Horizons* 10(6):1956–1968
 187. Sharma A, Naskar S, Mukhopadhyay T (2025) Programmable constitutive mode-coupling in tubular origami metamaterials, *npj metamaterials*
 188. Ghuku S, Naskar S, Mukhopadhyay T (2025) Stimuli-responsive programmable mechanics of bi-level architected nonlinear mechanical metamaterials. *Mech Mater* 206:105333
 189. Mondal S, Mukhopadhyay T, Naskar S (2025) Active heterogeneous mode coupling in bi-level multi-physically architected metamaterials for temporal, on-demand and tunable programming. *Commun Eng* 4(103)
 190. Tiwari P, Naskar S, Mukhopadhyay T (2025) Nonlinear functionally graded metamaterials for hydrogen storage and enhanced sustainability under extreme environments. *Thin-Walled Struct* 210:112901
 191. Sinha P, du D K, Naskar S, Mukhopadhyay T (2025) Effective elastic properties of 3D lattice materials with intrinsic stresses: bottom-up spectral characterization and constitutive programming. *Appl Math Modell* 140 115786
 192. Zheng L, Kochmann DM, Kumar S (2024) HyperCAN: hypernetwork-driven deep parameterized constitutive models for metamaterials. *Extrem Mech Lett* 72:102243
 193. Deng W, Kumar S, Vallone A, Kochmann DM, Greer JR (2024) AI-Enabled materials design of non-periodic 3D architectures with predictable direction-dependent elastic properties. *Adv Mater* 36(34):2308149
 194. Mukhopadhyay T, Adhikari S, Alu A (2019) Probing the frequency-dependent elastic moduli of lattice materials. *Acta Materialia* 165:654–665

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.